

Finite-alphabet transfinite words

Exact order types and guarded periodic blocks

A proposed solution to Altman's finite-alphabet sharpness question

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Abstract

For a finite poset P , consider all words of ordinal length less than ω^2 , ordered by strictly increasing subsequence embeddings that may increase labels in P . Write $n = |P|$ and let $j(P)$ count all downsets of P , including the empty one. This article gives a complete proposed proof that

$$o(s_{\omega^2}^F(P)) = \begin{cases} 1, & n = 0, \\ \omega^2, & n = 1, \\ \omega^{\omega^{n+j(P)-2}}, & n \geq 2. \end{cases}$$

In particular, the two-letter antichain has maximal order type ω^{ω^4} , answering the explicit sharpness question in Example 3.15 of Altman's March 2026 revision. The two-letter chain has type ω^{ω^3} . The main construction embeds every finite Cartesian power of an earlier stage into the next stage by synchronizing periodic ω -blocks. Proper downsets admit an excluded-letter guard; the universal block instead requires a limit-length padding argument. A more general classification for selected families of periodic tails isolates the exact exception in which naive block counting fails. Two routes from the product embeddings to the ordinal lower bound are given: the classical natural-product theorem, and an elementary hand-built linear extension of a Cartesian power that makes the main theorem independent of that import. Supplementary exact-length strata are computed, and they show explicitly why the answer cannot be assembled length by length. An exact symbolic embedding algorithm with canonical normal forms, negative controls for three invalid shortcuts, and two separately recorded computational suites accompany the proofs. The finite-height extension also computes the full word order below every ω^k , for positive finite k , from canonical finite atom posets.

Keywords. Well-quasi-order; maximal order type; transfinite word; recurrent downset; guarded periodic block; Higman ordering; finite downset lattice; ordinal arithmetic; natural product.

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1 The selected question and the result

Status and scope. This is a new, unrefereed research argument, not a report of an independently certified result, and it has not been checked in a proof assistant. All proof steps beyond the explicitly stated classical maximal-order-type theorems are supplied below. The specific unresolved question was checked in the cited source; the claim of a solution rests on the argument given here and not on a claim of exhaustive literature coverage. The computer checks test symbolic embeddings, not infinite ordinal ranks. The theorems concern *strictly increasing* embeddings and finite alphabet posets. The selected-family and exact-stratum results concern the strict cutoff ω^2 ; the full-language theorem also treats every finite height. Neither result settles arbitrary alphabets or the all-height union.

Provenance. This article consolidates three source manuscripts. The original two-source part merged reports on the same question, *order-types-below-omega-squared* and *guarded-periodic-blocks*. Both reached the same formula by the same core mechanism: a finite generating alphabet for the upper bound, cofinal synchronization of periodic separators, an excluded-letter guard for proper downsets, and a limit-length guard for the universal one. That shared core is proved exactly once below. The two reports diverged at one step, the passage from the Cartesian-power embeddings to the ordinal lower bound, and both arguments are kept: the classical natural-product route in Corollaries 5.3 and 6.4, and the elementary route of Section 6.4. Two different justifications of the monotone-surjection principle are likewise both recorded in Section 2.2. Material contributed by the second report, and absent from the first, includes the exact-length strata of Section 20, the canonical normal forms of Section 21.3, and the worked negative controls of Section 8.4. The third manuscript, formerly *finite-alphabet-transfinite-words*, contributes the finite-height induction and its separate implementation; see Section 1.2.

Altman studies finite-image words of bounded transfinite length in [1]. Example 3.15, on printed page 10 of version 2, gives the upper bound ω^{ω^4} for a two-element antichain and explicitly leaves its sharpness unverified. The same example improves the chain's upper bound to ω^{ω^3} . The arXiv record checked on September 19, 2026 listed version 2, dated March 10, 2026, as the latest revision. Targeted searches at that time did not locate a subsequent resolution; this is a limited literature check, not a guarantee of priority. No claim is made that the present literature check excludes every earlier formulation of the general finite-poset formula.

Related work. The classical maximal-linear-extension theory is due to de Jongh and Parikh [2], and the finite-word calculation is part of the de Jongh–Parikh–Schmidt theory [2, 3]; the underlying well-quasi-order assertion is Higman's lemma [4]. Recent work of Chopra and Pakhomov [5] studies the limit cutoff ω^ω through finite trees. The present article addresses the finite cutoffs ω^k and the explicitly identified two-letter example at $k = 2$; it does not settle the limit cutoff.

This is an attractive problem to attack because the alphabet is tiny, all ω -tails have finite descriptions up to embedding, and the candidate ordinal is explicit. The difficult part is not finding more tail types. It is proving that the types already available contribute their full ordinal complexity after concatenation.

Let P be a finite poset. A *downset* is a subset $D \subseteq P$ such that $x \leq_P y \in D$ implies $x \in D$. Put

$$\mathcal{J}(P) = \{D \subseteq P : D \text{ is a downset}\}, \quad j(P) = |\mathcal{J}(P)|.$$

The empty set is counted in $j(P)$ throughout.

Theorem 1.1 (Finite-alphabet formula). *Let P be a finite poset with $n = |P|$ elements. With strictly increasing, label-respecting subsequence embedding and mutual embeddability quotiented out,*

$$o(s_{\omega^2}^F(P)) = \begin{cases} 1, & n = 0, \\ \omega^2, & n = 1, \\ \omega^{\omega^{n+j(P)-2}}, & n \geq 2. \end{cases} \quad (1.1)$$

The two small alphabet sizes are part of the theorem, not removable technicalities. In particular, substituting $n = 1$ into the third line would give an incorrect answer; Section 7.2 explains why the defect is structural.

Here $s_{\omega^2}^F(P)$ denotes words of length *strictly less than* ω^2 . Since P is finite, the finite-image superscript is automatic. The theorem is proved in Section 7 after two product embedding lemmas.

Corollary 1.2 (The selected binary cases). *For a two-element antichain A_2 and a two-element chain C_2 ,*

$$o(s_{\omega^2}^F(A_2)) = \omega^{\omega^4}, \quad o(s_{\omega^2}^F(C_2)) = \omega^{\omega^3}.$$

Proof. There are four downsets in A_2 and three in C_2 . Substitute $n = 2$ in (1.1). $\square$

The proof establishes more than these two values. We may choose an arbitrary family of nonempty downsets and allow only the corresponding periodic ω -blocks. Unless the universal block is the sole allowed infinite block, the number of allowed block types determines the maximal order type exactly. The exceptional case is essential, not a technical inconvenience.

1.1 What the proof adds

The upper bound follows from a finite generating alphabet. The lower bound requires an additional construction. A new periodic block serves as a separator only when an embedding cannot hide it inside earlier blocks. Introducing downsets in inclusion order solves that problem. A finite guard then prevents a component from leaking into the next separator. At the universal separator no excluded letter exists, so a different device is needed: attach a cancellable tail of limit length. An infinite tail cannot fit into a bounded, hence finite, initial segment of an ω -block.

Together these observations give actual order embeddings

$$X^{r+1} \hookrightarrow Y \quad (r < \omega)$$

from each stage into the next. Classical natural-product arithmetic turns these embeddings into the sharp lower bound. Section 6.4 gives a second, elementary passage from the same embeddings to the same bound, using only an explicitly constructed linear extension of a Cartesian power; the main theorem therefore does not depend on importing the de Jongh–Parikh product formula.

What is claimed as the contribution. The normal-form encoding and the finite Higman maximal-order-type formula are established ingredients. The central contribution is the guarded product construction of Sections 5 and 6, which makes the generator-count upper bound sharp. The support-family classification and the finite-poset formula follow from that construction. The exact-length calculations of Section 20 are supplementary consequences, not the foundation of the main lower bound.

1.2 The finite-height extension and how the reports fit together

The third source, *Exact maximal order types of finite-alphabet transfinite words*, was also prepared on 19 September 2026. Its full-language formula at height two is precisely Theorem 1.1; its binary values and extremal-alphabet characterization are Corollaries 1.2 and 9.1. Those common statements are not repeated. The third source additionally treats every positive finite height. In contrast, the selected-family classification, exact-length strata and canonical token normalization below remain separate contributions of the two-source report. Neither contains the other in its full scope.

To prevent a collision, write $\mathcal{V}_k(P)$ for all words of length less than ω^k , including the empty word, and reserve $\mathcal{W}(P, \mathcal{F})$ for the selected periodic-tail families. Thus $\mathcal{V}_2(P) = s_{\omega^2}^F(P)$ up to the same mutual-embedding quotient. The third source wrote $\mathcal{W}_k$ for $\mathcal{V}_k$. Its $J(P)$ and the earlier $j(P)$ both count all downsets, including the empty one. Its periodic node $\Omega(S)$ repeats maximal generators; at height one it is equivalent to the all-elements representative $B_{\downarrow S}$. Natural and ordinary powers retain their distinct meanings.

The following theorem is proved in Sections 11–15. The height-uniform proof uses protected final segments and product layers; the earlier proof uses finite guards for proper downsets and limit padding for the universal downset. The elementary alternative in Section 6.4 removes the product-theorem import only from the height-two main result, not from the displayed higher-height proof or the exact-stratum calculus. All three manuscripts remain unrefereed and unformalized; consolidation does not establish priority or refresh their September 19 literature searches.

Precise definitions follow in [sections 2 and 11](#). For now, $\mathcal{I}_k(P)$ denotes the poset of equivalence classes of nonempty indecomposable P -words of lengths

$$1, \omega, \omega^2, \dots, \omega^{k-1},$$

and $N_k(P) = |\mathcal{I}_k(P)|$. A word is *indecomposable* here when it is mutually embeddable with every nonempty final segment of itself.

Theorem 1.3 (Finite-alphabet exactness). *Let P be a finite poset with $n \geq 2$ elements, and let $k \geq 1$ be finite. Then $\mathcal{I}_k(P)$ is finite and*

$$o(\mathcal{V}_k(P)) = \omega^{\omega^{N_k(P)-1}}. \quad (1.2)$$

Writing $J(R)$ for the number of order ideals of a finite poset R , including the empty ideal, the numbers are computed by

$$\mathcal{I}_0(P) = \emptyset, \quad N_1(P) = n, \quad N_{k+1}(P) = n + J(\mathcal{I}_k(P)) - 1. \quad (1.3)$$

The recurrence requires the actual order on $\mathcal{I}_k(P)$, not just its cardinality.

The full proof is in [section 15](#). Two consequences can already be stated.

Corollary 1.4 (Necessary exceptional cases). *For the empty alphabet, $o(\mathcal{V}_k(\emptyset)) = 1$. For the one-element alphabet,*

$$o(\mathcal{V}_k(1)) = \omega^k.$$

A finite quasi-order is first replaced by its quotient under mutual comparison; the number of classes, not the number of original labels, determines whether these exceptions apply.

Proof. There is only the empty word over the empty alphabet. Over one letter, a word is determined by its length up to equivalence, and embedding is exactly ordinal comparison of lengths. The set of lengths less than ω^k has order type ω^k . $\square$

2 Definitions and classical inputs

2.1 Ordinal words and embeddability

A word over P is a function $u : \alpha \rightarrow P$, where α is an ordinal; its length is $\text{len}(u) = \alpha$. If $u : \alpha \rightarrow P$ and $v : \beta \rightarrow P$, write $u \preceq v$ when there is a strictly increasing map $f : \alpha \rightarrow \beta$ satisfying

$$u(\xi) \leq_P v(f(\xi)) \quad (\xi < \alpha).$$

No continuity or cofinality condition is imposed on f . Write $u \simeq v$ when $u \preceq v$ and $v \preceq u$. Concatenation is ordinal concatenation. The empty word ε has length zero. For lengths below ω^2 there are unique $k, m < \omega$ such that

$$\text{len}(u) = \omega k + m.$$

Embeddability is in general only a quasi-order. Distinct words, such as a^ω and aa^ω , may be equivalent. All maximal order types in this article refer to the quotient by $\simeq$.

Strictness of the index map is indispensable. Under a merely nondecreasing index map an infinite constant source could be collapsed onto a single target position, and every statement below would describe a different problem. The same remark explains why finite material preceding an infinite block can disappear from the *length* expression, by $k + \omega = \omega$, while its labels may still matter for embeddability.

Lemma 2.1 (Concatenation is monotone). *If $u_i \preceq v_i$ for $0 \leq i < r$, where r is finite, then $u_0 \cdots u_{r-1} \preceq v_0 \cdots v_{r-1}$.*

Proof. Translate the embedding on each component into the corresponding interval of the target concatenation. The intervals occur in the same order, so the union of these translated maps is strictly increasing and respects labels. $\square$

A finite concatenation of words of length below ω^2 still has length below ω^2 . In particular, all maps used later stay within the stated cutoff, even though the number of ω -blocks in their arguments is unbounded across different arguments.

2.2 Maximal order types

A quasi-order is a *well-quasi-order* if every infinite sequence $x_0, x_1, \dots$ contains $i < j$ with $x_i \leq x_j$. The maximal order type $o(X)$ is the largest ordinal type of a linear extension of the quotient poset. The existence of this maximum, its characterization by the tree of bad finite sequences, and the product theorem below are classical [2]. We use the following precise inputs.

Theorem 2.2 (Classical maximal-order-type calculus). *For well-quasi-orders X, Y :*

- (i) *An order embedding $X \hookrightarrow Y$ implies $o(X) \leq o(Y)$.*
- (ii) *An order-preserving surjection from Y onto X , including surjectivity up to quasi-order equivalence, implies $o(X) \leq o(Y)$.*
- (iii) *For the componentwise Cartesian product, $o(X \times Y) = o(X) \otimes o(Y)$, where $\otimes$ is natural product.*
- (iv) *For a disjoint union, with no comparisons across the two parts, $o(X \amalg Y) = o(X) \oplus o(Y)$, where $\oplus$ is natural sum.*
- (v) *If P is a nonempty finite poset of cardinality n , its Higman word ordering satisfies*

$$o(P^*) = H_n, \quad H_n := \omega^{\omega^{n-1}}. \quad (2.1)$$

The finite-word formula is part of the de Jongh–Parikh–Schmidt theory [2, 3]; its underlying well-quasi-order assertion is Higman’s lemma [4]. These are imported theorems; the present article does not claim new proofs of them. All later lower bounds are reduced to these inputs by explicit embeddings. Assertions (iii), (iv) and (v) are used as follows: (v) everywhere, (iii) in the main line of Corollaries 5.3 and 6.4 and, together with (iv), in the exact-length calculations of Section 20. The alternative route of Section 6.4 shows that (iii) can be dropped from the main theorem, though not from Section 20.

Assertion (ii) admits two elementary explanations, and both are recorded because they illuminate different sides of the statement.

First argument, on bad sequences. Choose a preimage for each element of the target quotient. If a target sequence is bad, its sequence of chosen preimages is bad, because any comparison in the source would project to a comparison in the target. This embeds the target bad-sequence tree into the source tree. The same argument shows the target is a well-quasi-order whenever the source is.

Second argument, on fibres. Choose a linear extension of the target X of type $o(X)$ and order the fibres of the surjection consecutively according to that extension, then linearize each fibre arbitrarily. Because the map is order preserving, no comparison of the source is violated by this arrangement: a source comparison projects to a target comparison, which is respected by the chosen ordering of fibres, and comparisons inside a single fibre are respected by its own linearization. The result is a linear extension of Y of type at least $o(X)$, so $o(X) \leq o(Y)$. The first argument also yields the well-quasi-order assertion; the second exhibits a concrete linear extension and is what the exact-length upper bounds of Section 20 reuse.

2.3 The ordinal arithmetic used here

For ordinals in Cantor normal form, natural sum $\oplus$ adds coefficients at equal exponents, and natural product $\otimes$ multiplies the resulting formal polynomials, using natural sum on exponents. In particular,

$$(\omega^\alpha) \otimes (\omega^\beta) = \omega^{\alpha \oplus \beta}.$$

Consequently, for finite $N \geq 1$ and $q \geq 1$,

$$H_N^{\otimes q} = \omega^{\omega^{N-1}q}, \quad (2.2)$$

where the multiplication by q on the right is ordinary ordinal multiplication by a positive integer. Continuity of ordinal exponentiation gives

$$\sup_{q \geq 1} H_N^{\otimes q} = \omega^{\sup_{q \geq 1} \omega^{N-1}q} = \omega^{\omega^N} = H_{N+1}. \quad (2.3)$$

Equation (2.3) is the entire ordinal calculation behind the main induction.

A companion identity, using *ordinary* ordinal exponentiation only, will be needed for the alternative route of Section 6.4: for $q \geq 1$,

$$\sup_{r \geq 1} H_q^r = \sup_{r \geq 1} \omega^{\omega^{q-1} \cdot r} = \omega^{\omega^q} = H_{q+1}. \quad (2.4)$$

Ordinary powers and natural powers must not be interchanged in general, and (2.2)–(2.3) and (2.4) are emphatically not typographical variants of one another. In (2.2) the exponent $\otimes q$ is a natural power of H_N , computed by natural multiplication of Cantor normal forms; in (2.4) the exponent r is an ordinary ordinal power, computed by iterated ordinary multiplication. The two displays happen to have the same supremum, which is exactly why the two routes to the lower bound reach the same ordinal, and each display states exactly the special case that its own route uses. Neither may be substituted for the other inside a proof.

3 Periodic tails and the finite generating alphabet

3.1 One periodic block for each nonempty downset

For a nonempty downset $D = \{d_0, \dots, d_{t-1}\}$, fix a periodic word

$$B_D = (d_0 \cdots d_{t-1})^\omega. \quad (3.1)$$

The chosen enumeration is immaterial up to $\simeq$: every element of D appears infinitely often in every such representative.

Lemma 3.1 (Comparison of periodic blocks). *For nonempty downsets D, E and a letter $a \in P$,*

$$a \preccurlyeq B_D \iff a \in D, \quad B_D \preccurlyeq B_E \iff D \subseteq E.$$

Every nonempty tail of B_D is equivalent to B_D , and a letter a embeds into every tail of B_D exactly when $a \in D$. Consequently, if every letter of a finite word p lies in D , then

$$pB_D \simeq B_D. \quad (3.2)$$

Proof. If $a \leq_P d$ for a letter $d \in D$, downward closure gives $a \in D$. Conversely, $a \in D$ supplies a matching occurrence in B_D . If $B_D \preccurlyeq B_E$, applying the same observation to an occurrence of each $d \in D$ gives $D \subseteq E$. If $D \subseteq E$, recursively match the letters of B_D to successive identical occurrences in B_E . There is always a later occurrence, so the recursion gives an embedding. The identical recursive construction works in any nonempty tail of a periodic block; the reverse embedding into the full block is inclusion. For (3.2), place the finitely many letters of p in an initial portion of B_D , which is possible because each of them lies in D and every element of D recurs, and embed the remaining copy of B_D into the tail that is left. The reverse embedding simply omits p . $\square$

Note that (3.2) holds for an arbitrary nonempty downset, not only for $D = P$. The special case $D = P$, where p may be any finite word at all, is isolated later as (6.1).

For a nonempty subset $R \subseteq P$, repeating R or repeating its downset $\downarrow R$ gives equivalent ω -words. In the less immediate direction, each $x \in \downarrow R$ can be matched to an element $r \in R$ above it, and each such r recurs infinitely often.

Remark 3.2 (Repeating only the maximal elements). Let $\text{Max } D$ denote the set of maximal elements of a nonempty downset D . Applying the previous paragraph with $R = \text{Max } D$, whose downset is D , gives

$$B_{\text{Max } D} \simeq B_D.$$

Some presentations of this material define the periodic representative by repeating $\text{Max } D$ rather than all of D . The two conventions are interchangeable everywhere below, and we keep the all-elements block B_D of (3.1) as the official definition. Where a proof below inspects the letters of a block, it only ever uses that the letters lie in D and that every element of D is dominated by a recurrent letter. Both representatives have this property; literal recurrence of every element of D holds only for the all-elements representative.

3.2 Reduction of arbitrary words

For a word $u : \omega \rightarrow P$ put

$$\text{Rec}(u) = \{a \in P : a \text{ occurs infinitely often in } u\}, \quad I(u) = \downarrow \text{Rec}(u). \quad (3.3)$$

The set $\text{Rec}(u)$ is nonempty because P is finite and u is infinite. Its downward closure $I(u)$, and not the literal recurrent set, is the correct invariant here: an embedding may increase labels, so what a block can absorb is determined by the downset generated by its recurrent letters. We call $I(u)$ the *recurrent downset* of u .

The following periodic-tail reduction is standard; compare [1, Propositions 3.3–3.4]. A proof is included to specify exactly which equivalence is needed.

Lemma 3.3 (Finite prefix and recurrent tail). *Every word $u : \omega \rightarrow P$ is equivalent to vB_D for some finite word $v \in P^*$ and some nonempty downset D ; one may take $D = I(u)$.*

Proof. Let $R = \text{Rec}(u)$ be the set of labels occurring infinitely often in u . Since P is finite and u is infinite, R is nonempty. Every label outside R occurs only finitely often. Choose an index after all such occurrences, and write $u = vt$, where v is finite and t uses only labels from R . Every element of R occurs infinitely often in t .

Put $D = \downarrow R$. To embed t into B_D , successively choose later identical occurrences of its labels. To embed B_D into t , for each requested label $d \in D$ choose an $r \in R$ with $d \leq_P r$, and then choose an occurrence of r later than all previous choices. Infinitely many occurrences of r make this possible. Thus $t \simeq B_D$, and monotonicity of concatenation gives $u \simeq vB_D$. $\square$

The prefix can be normalized further, and the strengthened form is what the canonical representatives of Section 21.3 and the exact-length computations of Section 20 require.

Lemma 3.4 (One-block normal form). *Every word $u : \omega \rightarrow P$ is equivalent to pB_D with $D = I(u)$, where the finite word p may in addition be required to be empty or to end in a letter outside D .*

Proof. Each letter outside D occurs only finitely often in u , and there are finitely many letters, so either no letter outside D occurs, in which case put $p = \varepsilon$, or there is a last such occurrence, in which case let p be the prefix ending at that occurrence. In both cases write $u = pt$ with every label of t inside D .

Every element of D is still dominated by a recurrent letter in t : given $d \in D$ there is, by $D = \downarrow \text{Rec}(u)$, a recurrent $b \geq_P d$, and only finitely many positions were removed with p , so b still recurs in t . Greedy matching of the labels of t to dominating occurrences gives $t \preceq B_D$, and matching each requested $d \in D$ to a later occurrence of such a b gives $B_D \preceq t$. Hence $t \simeq B_D$, and concatenation gives $u \simeq pB_D$. $\square$

Proposition 3.5 (Finite representation). *Every word of length less than ω^2 is equivalent to a finite concatenation of one-letter words and blocks B_D , with $\emptyset \neq D \in \mathcal{J}(P)$. More precisely, a word of length $\omega r + k$ is equivalent to an expression*

$$p_0 B_{D_0} p_1 B_{D_1} \cdots p_{r-1} B_{D_{r-1}} p_r, \quad (3.4)$$

in which every p_i is a finite word, every p_i with $i < r$ is empty or ends in a letter outside D_i , and p_r has length k .

Proof. Split a word of length $\omega r + k$ into its r consecutive canonical intervals of length ω and its final finite interval of length k . Apply Lemma 3.3 to each infinite interval and concatenate the resulting representatives; applying Lemma 3.4 instead yields the normalized prefixes of (3.4). $\square$

By itself this is a representation theorem, not a uniqueness theorem: without the extra clause in (3.4), different finite expressions represent equivalent words, for example $aB_{\downarrow a} \simeq B_{\downarrow a}$. Such absorption is precisely why a finite generating alphabet gives only an upper bound without further work. Uniqueness is nevertheless recoverable once the prefixes are normalized as in (3.4); this is carried out, with an explicit stack-based procedure, in Section 21.3.

The construction is existential for an arbitrary infinite input word, since it uses knowledge of which letters recur. The algorithm of Section 21 takes finite expressions as input; it is not an algorithm for discovering recurrence in an arbitrary stream.

Definition 3.6 (Selected families). For $\mathcal{F} \subseteq \mathcal{J}(P) \setminus \{\emptyset\}$, let $\mathcal{W}(P, \mathcal{F})$ be the quasi-order of all finite concatenations of letters in P and periodic blocks B_D with $D \in \mathcal{F}$, using actual ordinal-word embeddability. Finite expressions are identified up to mutual embeddability when maximal order type is taken.

In particular, $\mathcal{W}(P, \emptyset) = P^*$, and $\mathcal{W}(P, \mathcal{J}(P) \setminus \{\emptyset\})$ has the same quotient order as $s_{\omega^2}^F(P)$ by Proposition 3.5.

Remark 3.7 (An intrinsic description of $\mathcal{W}(P, \mathcal{F})$). Definition 3.6 is syntactic, but the resulting class of words has an intrinsic description: up to equivalence, $\mathcal{W}(P, \mathcal{F})$ consists exactly of those words below ω^2 each of whose canonical ω -blocks has its recurrent downset in $\mathcal{F}$. A finite prefix cannot change the recurrent downset of a block, which gives one inclusion, and Lemma 3.4 gives the converse by rewriting each block as pB_D with $D = I(u) \in \mathcal{F}$.

Proposition 3.8 (Atom-count upper bound). For $n = |P| \geq 1$ and $m = |\mathcal{F}|$,

$$o(\mathcal{W}(P, \mathcal{F})) \leq H_{n+m}. \quad (3.5)$$

Moreover, $\mathcal{W}(P, \mathcal{F})$ is a well-quasi-order.

Proof. Take a discrete alphabet of $n + m$ symbols: one for each finite letter and one for each selected periodic block. Evaluate a finite symbol string by ordinal concatenation. If one string is a subsequence of another, the corresponding evaluated word embeds into the other by Lemma 2.1. Evaluation is surjective onto the represented quasi-order. Apply the finite-word formula and monotone-surjection principle in Theorem 2.2. $\square$

Using a discrete alphabet deliberately ignores extra comparisons among the atoms. This does not weaken the desired upper bound: every poset with the same positive finite cardinality has the same finite-word maximal order type. No order-reflection claim is made for the evaluation map.

4 Cofinal localization and synchronization

The following lemma concerns *displayed target tokens*. A finite word may contain many one-letter tokens adjacent to each other, and a token expression need not be a canonical decomposition of its ordinal length. Nevertheless, each displayed B_E occupies a convex interval of order type ω , and there are only finitely many displayed tokens.

Lemma 4.1 (Cofinal localization). Suppose a periodic block B_D embeds into a finite token expression v . Then a tail of its image lies in one displayed target block B_E , is cofinal in that block, and satisfies $D \subseteq E$. If a source position occurs after the whole source block B_D , its image occurs after the whole target block B_E .

Proof. Number the target tokens in their order of occurrence. As the source positions $0, 1, 2, \dots$ increase, the indices of the tokens containing their images form a nondecreasing sequence with only finitely many possible values. It is eventually constant. The final token cannot be a one-letter token, because the embedding is injective. It is therefore some B_E . The image contains infinitely many positions of that token, so it is unbounded in its order type ω .

Every $d \in D$ still occurs in the source tail. Each such occurrence maps to a label in E above d . Downward closure gives $d \in E$, proving $D \subseteq E$. Finally, the image of any subsequent source position must lie above every image point of B_D . Since those points are cofinal in the target B_E , no position of that target block remains available. $\square$

Remark 4.2 (A block may straddle earlier blocks). The lemma does not say that the entire source block lands in a single target block. An initial finite part can be placed earlier. Only the tail is localized. All subsequent proofs use this weaker, correct statement. Under the stronger hypothesis that source and target have the *same* number of canonical ω -blocks, whole-block containment does hold; that is Lemma 20.2 below, which is used only in Sections 20 and 21.3. The two statements are compatible because their hypotheses differ, and nothing in Lemma 20.2 retracts the present remark.

Remark 4.3 (The role of explicit segments). A displayed token B_E inside an expression need not begin at a canonical ω -block boundary of the word it denotes: if a finite word precedes it, its pure periodic segment starts at a finite offset inside the corresponding canonical block. That segment still has length ω , is cofinal in the block, and ends at the next canonical boundary; this is the ordinal identity $k + \omega = \omega$ again. Throughout the arguments below, “inside a displayed separator” therefore always means inside that *explicit pure periodic segment*, never inside the finite prefix that happens to precede it. This is a statement about the *target* side, complementary to Remark 4.2, which is about a *source* block straddling earlier target blocks.

Lemma 4.4 (Equal-count synchronization). *Let D be a nonempty downset, and suppose an old family $\mathcal{G}$ satisfies*

$$D \not\subseteq E \quad \text{for every } E \in \mathcal{G}. \quad (4.1)$$

Consider two token expressions with exactly r displayed copies of B_D , all other infinite tokens belonging to $\mathcal{G}$. Under an embedding of the first expression into the second, the tail of the i th displayed source B_D is cofinal in the i th displayed target B_D , for $1 \leq i \leq r$.

Proof. By Lemma 4.1, each displayed source B_D requires a target infinite block whose downset contains D . Condition (4.1) excludes all old target blocks. Its cofinal destination is therefore one of the r displayed new target blocks. Successive source new blocks have strictly increasing cofinal destination indices: after the first has used a target block cofinally, any later source position must be beyond it. A strictly increasing injection from $\{1, \dots, r\}$ into itself is the identity. This proves the claim. $\square$

Corollary 4.5 (Location between synchronized boundaries). *In the situation of Lemma 4.4, write the source expression as $b_0 B_D b_1 B_D \cdots B_D b_r$ and the target as $d_0 B_D d_1 B_D \cdots B_D d_r$, with all b_i, d_i in $\mathcal{W}(P, \mathcal{G})$. Then, for each i :*

- (a) *if $i > 0$, every image point of b_i occurs at or after the end of the i th displayed target separator;*
- (b) *if $i < r$, every image point of b_i occurs before the image of the first point of the $(i + 1)$ st source separator, hence before the end of the $(i + 1)$ st displayed target separator;*
- (c) *b_r embeds into d_r .*

Proof. Every point of b_i follows every point of the preceding source separator. That separator’s image is cofinal in the corresponding target separator by Lemma 4.4, so no target position of that separator remains available, giving (a). Every point of b_i precedes the next source separator, whose first image point lies before the end of the corresponding target separator, again by cofinality; strict monotonicity gives (b). After the last target separator, the only remaining target material is d_r , which with (a) gives (c). $\square$

Part (b) still permits a finite tail of b_i to enter the next target separator. The guards of Sections 5 and 6 eliminate precisely that possibility.

To arrange (4.1), introduce a finite collection of downsets in an inclusion-respecting order. Ordering them by cardinality, with arbitrary tie breaking, suffices: a new set cannot be contained in an earlier, distinct set of no larger size. The full downset P , when present, is last.

5 Proper markers: finite guards give product embeddings

Lemma 5.1 (Right cancellation of a final letter). *For arbitrary ordinal words u, v and any fixed letter c ,*

$$uc \preccurlyeq vc \implies u \preccurlyeq v.$$

Proof. The image of the final source letter lies at or before the final target letter. Every position belonging to u maps strictly before that image, hence inside the target prefix v . Restrict the embedding. $\square$

Theorem 5.2 (Proper-marker product embedding). *Let D be a nonempty proper downset of P , and let $\mathcal{G}$ satisfy (4.1). Choose $c \in P \setminus D$ and put $X = \mathcal{W}(P, \mathcal{G})$, $Y = \mathcal{W}(P, \mathcal{G} \cup \{D\})$. For every $r \geq 1$, the map*

$$\Phi_r(u_0, \dots, u_r) = u_0 c B_D u_1 c B_D \cdots u_{r-1} c B_D u_r \quad (5.1)$$

is an order embedding $X^{r+1} \hookrightarrow Y$.

Proof. Order preservation follows by concatenating componentwise embeddings and identity maps on the separators. It remains to prove reflection.

Suppose $\Phi_r(u_0, \dots, u_r) \preceq \Phi_r(v_0, \dots, v_r)$. By Lemma 4.4, the i th source separator B_D has image tail cofinal in the i th target separator B_D . Thus, for $i > 0$, every image point of u_i and of its subsequent guard lies after the i th target separator. For $i = 0$ there is no preceding separator and no lower-bound restriction to impose.

Fix $i < r$. The source guard c immediately following u_i precedes the next source separator. Some point of that next separator maps into the corresponding next target separator. Therefore the source guard's image is before the end of that target separator. It cannot be in that separator: if it were matched to a label $d \in D$, then $c \leq_P d$ would imply $c \in D$, contrary to the choice of c . Its image consequently lies in the target interval $v_i c$ before the separator. Every point of u_i precedes the guard, so its image lies in v_i , not in the final guard of $v_i c$. We obtain $u_i \preceq v_i$.

After the last source separator, cofinality forces the entire source suffix u_r to map into the target suffix v_r . Hence $u_r \preceq v_r$ as well. All components compare in the product order. Reflection and preservation imply an order embedding after taking equivalence classes. $\square$

The guard is allowed to map into the interior of v_i ; it need not match the explicitly displayed target guard. The proof only needs the fact that it lies no later than that guard. This is important when old components already contain many occurrences of c or old infinite blocks.

Corollary 5.3 (One proper marker raises the level). *Under the hypotheses of Theorem 5.2, if $o(X) = H_N$ and $n + |\mathcal{G}| = N$, then $o(Y) = H_{N+1}$.*

Proof. For every $r \geq 1$, product embedding and the classical product theorem give $o(Y) \geq H_N^{\otimes(r+1)}$. Taking the supremum and using (2.3) yields $o(Y) \geq H_{N+1}$. Proposition 3.8 gives the reverse inequality. $\square$

6 The universal marker: limit padding replaces a guard

For $D = P$ there is no excluded letter. Any fixed finite word z is absorbed by the universal periodic block:

$$zB_P \simeq B_P. \quad (6.1)$$

Indeed, the countable sequence zB_P can be matched greedily into B_P ; every requested label reappears infinitely often. Thus attaching a finite guard before B_P cannot preserve arbitrary component information. The remedy is to end the component with a cancellable *infinite* tail.

6.1 A cancellable map into limit lengths

Lemma 6.1 (Limit padding). *Let E be a nonempty proper downset of P , and choose $c \notin E$. The map*

$$G(u) = ucB_E \quad (6.2)$$

is order preserving and order reflecting on arbitrary words of length below ω^2 . Every $G(u)$ has nonzero limit length, of the form ωk with $k \geq 1$. If B_E is allowed in a family, G maps that family's represented word order into itself.

Proof. Monotonicity is immediate. Suppose $ucB_E \preceq vcB_E$. The source occurrence of c cannot map into the final target B_E , by downward closure and $c \notin E$. Thus it maps into vc , and the restriction to u maps into v , by the same final-letter argument as Lemma 5.1. Hence $u \preceq v$.

If $\text{len}(u) = \omega k + m$, then

$$\text{len}(ucB_E) = (\omega k + m) + 1 + \omega = \omega(k + 1),$$

which is a nonzero limit ordinal. The family assertion follows from the explicit concatenation. $\square$

Lemma 6.2 (No infinite tail fits below a finite bound). *Let u have nonzero limit length below ω^2 . Suppose an embedding maps all of u to positions preceding a fixed position η inside a target interval of order type ω . Then no position of u can map inside that interval.*

Proof. If a source position ξ mapped into the interval before η , there would be only finitely many target positions between its image and η . But a nonzero limit ordinal has infinitely many positions after each of its positions: otherwise the ordinal would have a last position. The infinitely many source positions after ξ would have to inject into that finite target interval. This is impossible. $\square$

The position η need not be the first position of the target ω -block. Any point of it gives a finite bound. Nor is it necessary for the next source marker to begin inside that block: one image point of its cofinal tail suffices.

The lemma must not be misread as saying that a word of limit length never embeds into a universal periodic word. It certainly may; B_P alone embeds into B_P . The obstruction exploited below is different, and it is created by the surrounding configuration: a further synchronized source separator is required to begin at an actual target position before the end of that same target block, so the padded component and that separator would have to share one interval of order type ω with only finitely many positions between them.

6.2 The product embedding

Theorem 6.3 (Universal-marker product embedding). *Let $\mathcal{G}$ consist of nonempty proper downsets of P , with $\mathcal{G} \neq \emptyset$. Choose $E \in \mathcal{G}$ and the map G from Lemma 6.1. Put $X = \mathcal{W}(P, \mathcal{G})$ and $Y = \mathcal{W}(P, \mathcal{G} \cup \{P\})$. For each $r \geq 1$,*

$$\Psi_r(u_0, \dots, u_r) = G(u_0)B_P G(u_1)B_P \cdots G(u_{r-1})B_P u_r \quad (6.3)$$

is an order embedding $X^{r+1} \hookrightarrow Y$.

Proof. Preservation follows from the monotonicity of G and concatenation. Suppose the image of $(u_0, \dots, u_r)$ embeds into the image of $(v_0, \dots, v_r)$.

No old downset contains P . Lemma 4.4 therefore applies to the r displayed universal separators. Their cofinal destinations are the corresponding r displayed target universal separators.

Fix $i < r$. When $i > 0$, cofinality at the preceding separator forces all of $G(u_i)$ to map after the corresponding preceding target separator. Choose an image point η of the following source B_P inside the corresponding following target B_P ; such a point exists by cofinal localization. Every image point of $G(u_i)$ is before η . By Lemma 6.1, $G(u_i)$ has nonzero limit length. Applying Lemma 6.2 shows that none of its positions can be in this following target B_P . Thus the whole image of $G(u_i)$ lies in the intervening target component $G(v_i)$. For $i = 0$ the same argument applies without a preceding boundary. We have proved

$$G(u_i) \preceq G(v_i) \quad (i < r).$$

Reflection for G yields $u_i \preceq v_i$ for all $i < r$.

After the last source universal separator, cofinality forces the final source component u_r into the final target component v_r . Hence $u_r \preceq v_r$, completing product reflection. $\square$

Corollary 6.4 (One universal marker raises the level when padding exists). *Under the hypotheses of Theorem 6.3, if $o(X) = H_N$ and $n + |\mathcal{G}| = N$, then $o(Y) = H_{N+1}$.*

Proof. Apply the finite Cartesian-product theorem to the embeddings Ψ_r , take the supremum in (2.3), and combine with the atom-count upper bound, exactly as in Corollary 5.3. $\square$

Notice the role of the old proper block. It need not encode the whole component by itself. Its job is to give a limit-length ending, while an excluded finite letter makes that ending cancellable. These two properties are logically distinct and both are used.

6.3 The two constructions as one statement

Theorems 5.2 and 6.3 differ only in which guard map is attached to a component. It is convenient, and it is what the quick reference of Appendix A states, to record them as a single theorem.

Theorem 6.5 (Unified guarded product embedding). *Let D be a nonempty downset not in $\mathcal{G}$ with $D \not\subseteq E$ for every $E \in \mathcal{G}$, and assume either*

- (i) $D \neq P$, or
- (ii) $D = P$ and $\mathcal{G}$ contains a nonempty proper downset.

Define a guard map g_D on $\mathcal{W}(P, \mathcal{G})$ by

$$g_D(u) = \begin{cases} uc, & D \neq P, c \in P \setminus D, \\ ucB_R, & D = P, R \in \mathcal{G} \text{ proper}, c \in P \setminus R, \end{cases} \quad (6.4)$$

and put, for $m \geq 0$,

$$\Phi_m(u_0, \dots, u_m) = g_D(u_0)B_D g_D(u_1)B_D \cdots B_D g_D(u_m). \quad (6.5)$$

Then

$$\Phi_m(u_0, \dots, u_m) \preceq \Phi_m(v_0, \dots, v_m) \iff u_i \preceq v_i \text{ for every } i \leq m,$$

so Φ_m is an order embedding $\mathcal{W}(P, \mathcal{G})^{m+1} \hookrightarrow \mathcal{W}(P, \mathcal{G} \cup \{D\})$.

Proof. Both guard maps are order preserving and order reflecting and send $\mathcal{W}(P, \mathcal{G})$ into itself: for case (i) this is Lemma 5.1 together with the concatenation lemma, and for case (ii) it is Lemma 6.1. Preservation of Φ_m follows by concatenating componentwise embeddings with identity maps on the separators.

For reflection, Lemma 4.4 aligns the m displayed separators and Corollary 4.5 places each guarded component between the corresponding target boundaries. In case (i) the terminal letter c of $g_D(u_i)$ cannot lie inside the next explicit target separator, because $c \notin D$ and D is downward closed, and it cannot lie after that separator by Corollary 4.5(b); so the whole of $g_D(u_i)$ lies in $g_D(v_i)$. In case (ii) the word $g_D(u_i)$ has nonzero limit length by Lemma 6.1, so Lemma 6.2 forbids it from meeting the next explicit target separator at all, with the same conclusion. The last component is handled by Corollary 4.5(c). Reflection of g_D then gives $u_i \preceq v_i$ for every i . $\square$

Remark 6.6 (Guarding the final component or not). Theorem 6.5 guards *every* component, including the last one, whereas the maps Φ_r of (5.1) and Ψ_r of (6.3) leave the final component u_r unguarded. Both conventions are correct and both are used in the literature on this construction. With a trailing guard the final component is handled by cancellation, exactly like the others; without one it is handled by Corollary 4.5(c), since nothing follows it in the target. We take the unguarded form as the official construction of Sections 5 and 6, because it is the shorter expression, and record the all-guarded variant here and in Appendix A so that the two displays are not mistaken for a discrepancy.

6.4 An elementary alternative to the product theorem

Corollaries 5.3 and 6.4 pass from the Cartesian-power embeddings to the ordinal lower bound through Theorem 2.2(iii), the de Jongh–Parikh natural-product formula. That import can be avoided. The following lemma supplies, by hand, the only consequence of it that the main theorem needs, and the route is recorded here in full because it changes the hypotheses of the main theorem rather than merely restating it.

Lemma 6.7 (Finite powers amplify a maximal order type). *For every well-quasi-order X and every finite $r \geq 1$,*

$$o(X^r) \geq o(X)^r, \quad (6.6)$$

with ordinary ordinal exponentiation on the right.

Proof. Let L be a maximal linear extension of the quotient of X , of type $\alpha = o(X)$. Order r -tuples by comparing their *last unequal coordinate*, using L in that coordinate. This is a linear order, and it is a well-order of type α^r : the last coordinate indexes α consecutive copies of the corresponding order on $(r-1)$ -tuples, so the type satisfies $\alpha_r = \alpha_{r-1} \cdot \alpha$ with $\alpha_1 = \alpha$.

It extends the componentwise order. Indeed, if one tuple is componentwise below another and the two differ, then at every coordinate the first is below or equal in L , and at the last coordinate where they differ it is strictly earlier. Hence the last-unequal-coordinate order is a linear extension of X^r , and (6.6) follows from the definition of o . $\square$

Corollary 6.8 (One new block raises one Higman level, elementarily). *Let $X = \mathcal{W}(P, \mathcal{G})$ and $Y = \mathcal{W}(P, \mathcal{G} \cup \{D\})$ be as in Theorem 6.5, and suppose $o(X) = H_N$ with $n + |\mathcal{G}| = N$. Then $o(Y) = H_{N+1}$.*

Proof. For every $r \geq 1$, Theorem 6.5 gives an order embedding $X^r \hookrightarrow Y$, so $o(Y) \geq o(X^r) \geq H_N^r$ by Theorem 2.2(i) and Lemma 6.7. Taking the supremum over r and applying the ordinary-exponentiation identity (2.4) gives $o(Y) \geq H_{N+1}$. Proposition 3.8, with one more atom, gives the reverse inequality. $\square$

Only an explicit linear extension of a Cartesian product was used here. In particular, the argument does not require any general formula for the maximal order type of a product, nor any formula for lexicographic products of arbitrary partial orders. This is what the alternative route buys: with Corollary 6.8 in place of Corollaries 5.3 and 6.4, the proof of Theorem 1.1 uses from Theorem 2.2 only the monotonicity assertions (i) and (ii) and the finite-word formula (v), and in particular no natural-product theorem. The main line of the article nevertheless keeps the classical route, both because it is the shorter derivation and because assertions (iii) and (iv) are genuinely needed later, in the exact-length calculations of Section 20, which the elementary bound (6.6) does not supply.

7 The full classification and the main theorem

Theorem 7.1 (Selected-tail classification). *Let P be a nonempty finite poset of size n , and let $\mathcal{F} \subseteq \mathcal{J}(P) \setminus \{\emptyset\}$.*

(i) *If $\mathcal{F} \neq \{P\}$, then*

$$o(\mathcal{W}(P, \mathcal{F})) = H_{n+|\mathcal{F}|} = \omega^{\omega^{n+|\mathcal{F}|-1}}. \quad (7.1)$$

This includes the empty family.

(ii) *If $\mathcal{F} = \{P\}$, then*

$$o(\mathcal{W}(P, \{P\})) = H_n \cdot \omega = \omega^{\omega^{n-1}+1}. \quad (7.2)$$

Proof of part (i). For the empty family, (7.1) is the finite-word formula. Otherwise $\mathcal{F} \neq \{P\}$ implies that the family contains at least one proper downset. List its proper downsets as $D_1, \dots, D_t$ in increasing cardinality, breaking ties arbitrarily. If $P \in \mathcal{F}$, append it last.

Start with $X_0 = P^*$, of type H_n . At step $i \leq t$, no earlier downset contains D_i . Indeed, an earlier downset has no greater cardinality, and equality together with containment would make the downsets equal. Theorem 5.2 applies. Inductively, Corollary 5.3, or equally Corollary 6.8 if one prefers the elementary route, gives

$$o(\mathcal{W}(P, \{D_1, \dots, D_i\})) = H_{n+i}.$$

If P is absent from $\mathcal{F}$, this finishes the proof. If P is present, there is an old proper downset because $t \geq 1$. Apply Theorem 6.3 and Corollary 6.4 to the final step, obtaining $H_{n+t+1} = H_{n+|\mathcal{F}|}$. $\square$

Proof of part (ii). Every expression using only the universal infinite block can be written

$$u_0 B_P u_1 B_P \cdots u_{k-1} B_P v, \quad u_i, v \in P^*.$$

By (6.1), it is equivalent to $B_P^k v$. Here B_P^k means k successive copies, not an infinite repetition. These normal forms satisfy

$$B_P^k v \preceq B_P^\ell w \iff (k < \ell) \text{ or } (k = \ell \text{ and } v \preceq w). \quad (7.3)$$

If $k < \ell$, send the k source infinite blocks into the first k target blocks and the finite suffix into one extra universal target block. If $k > \ell$, k successive source infinite blocks require k distinct cofinal target infinite destinations by Lemma 4.1, which is impossible. If $k = \ell$, synchronization leaves the source suffix wholly in the target suffix, so $v \preceq w$ is necessary. It is plainly sufficient.

Thus the quotient is the ordered sum of ω copies of P^* :

$$\mathcal{W}(P, \{P\}) / \approx \cong \sum_{k < \omega} P^*. \quad (7.4)$$

Every linear extension must put all of the k th copy before the $(k+1)$ st copy. Each copy has maximal order type H_n , and one may use a maximal linear extension in every copy. Therefore the maximal order type is the ordinary ordered sum $\sum_{k < \omega} H_n = H_n \cdot \omega$. The elementary identity $\omega^\alpha \cdot \omega = \omega^{\alpha+1}$ gives (7.2). $\square$

Proof of Theorem 1.1. Suppose $n \geq 2$. Choose a minimal element a of P . The singleton $\{a\}$ is a nonempty proper downset. Consequently the full family $\mathcal{J}(P) \setminus \{\emptyset\}$ is not the exceptional family $\{P\}$. Its size is $j(P) - 1$. Proposition 3.5 and Theorem 7.1(i) give

$$o(s_{\omega^2}^F(P)) = H_{n+j(P)-1} = \omega^{\omega^{n+j(P)-2}}.$$

For $n = 1$, there is one word at each ordinal length $\alpha < \omega^2$, and embedding is exactly comparison of lengths. Its quotient is therefore the ordinal ω^2 itself. Equivalently, apply part (ii) with $H_1 = \omega$. For $n = 0$, the empty word is the sole word, giving maximal order type 1. $\square$

7.1 Why the supremum argument is legitimate

The proof never identifies the order type of an increasing union with the supremum of the order types of its pieces. Such an identification is not a general principle for maximal order types. Instead, for each fixed finite r , there is a separately verified order embedding of a Cartesian power. It gives a genuine lower bound on the *same* target order type. Taking the supremum of a collection of lower bounds is valid. The upper bound is independently obtained by a monotone surjection.

The distinction is not pedantic, and Section 20 makes it concrete. There the maximal order type of each individual length stratum is computed exactly, and the supremum of those values, recorded in (20.7), is ω^{ω^n} : strictly smaller than the answer $\omega^{\omega^{n+j(P)-2}}$ of Theorem 1.1 whenever $n \geq 2$. A stratum-by-stratum supremum therefore genuinely fails to compute the theorem, while the supremum of lower bounds used here does not.

7.2 The singleton warning is structural

For a singleton alphabet, two formal atoms are available: the letter and its periodic ω -block. Their finite words would have maximal type $H_2 = \omega^\omega$ if treated as a free Higman alphabet. But after ordinal evaluation, the actual type is only ω^2 . The family $\mathcal{F} = \{P\}$ exhibits the same defect over every nonempty finite alphabet. Hence atom counting alone cannot be a lower-bound proof. The padding construction explains exactly why the full family over $n \geq 2$ avoids this collapse.

8 The binary proof written out explicitly

This section specializes the general proof so the selected problem can be checked without tracking an arbitrary downset family.

8.1 Two incomparable letters

Let $P = \{a, b\}$ with no comparisons between distinct letters, and abbreviate

$$A = a^\omega, \quad B = b^\omega, \quad U = (ab)^\omega.$$

There are exactly three nonempty downsets, giving these three periodic block types. Define stages by their allowed tokens, and record the guard used to activate the block that is new in each row:

stage	allowed tokens	guard for the new block	claimed maximal type
X_0	a, b	none; finite words	$H_2 = \omega^\omega$
X_1	a, b, A	$u \mapsto ub$	$H_3 = \omega^{\omega^2}$
X_2	a, b, A, B	$u \mapsto ua$	$H_4 = \omega^{\omega^3}$
X_3	a, b, A, B, U	$u \mapsto uba^\omega$	$H_5 = \omega^{\omega^4}$

The upper bound in each row comes from the number of tokens. The guard in the last row is the limit padding G of Lemma 6.1; the guards in the two middle rows are single excluded letters. The first equality is classical. To pass from X_0 to X_1 , use, for every finite r ,

$$(u_0, \dots, u_r) \mapsto u_0 b A u_1 b A \cdots u_{r-1} b A u_r.$$

Source A -blocks must synchronize with target A -blocks, and the guard b cannot enter A . This embeds X_0^{r+1} .

To pass from X_1 to X_2 , use

$$(u_0, \dots, u_r) \mapsto u_0 a B u_1 a B \cdots u_{r-1} a B u_r.$$

A source B -block cannot be cofinal in an old target A -block, and its preceding guard a cannot enter a target B -block. This embeds X_1^{r+1} .

For the final step, set

$$G(u) = u b A.$$

The guard b cannot enter the final A , so G reflects embedding; the image has limit length. Now use

$$(u_0, \dots, u_r) \mapsto G(u_0) U G(u_1) U \cdots G(u_{r-1}) U u_r.$$

A source U contains both letters infinitely often, so neither A nor B can be its cofinal destination. The displayed U -blocks synchronize. Limit padding prevents the intervening components from entering the next U before a fixed image point of its source marker. This embeds X_2^{r+1} .

At each of the three steps, (2.3) converts these product embeddings into the next displayed ordinal. Finally, every binary word of length below ω^2 is equivalent to an expression in a, b, A, B, U . This proves the sharpness of the binary antichain bound.

8.2 Two ordered letters

Let $0 < 1$. The nonempty downsets are $\{0\}$ and $\{0, 1\}$; their periodic blocks may be represented by

$$A = 0^\omega, \quad U = 1^\omega.$$

The latter is equivalent to $(01)^\omega$. Begin with finite words over $\{0, 1\}$, of maximal type H_2 . Add A using the guard 1, giving H_3 . Then add U using the limit padding $G(u) = u1A$, giving H_4 . Thus the chain's sharpened upper bound is also attained.

8.3 Three easily confused comparisons

For the antichain, the following facts illustrate the mechanism:

$$aU \simeq U, \quad A \preceq U, \quad AB \not\preceq U.$$

The first is finite-prefix absorption. The second uses cofinal recurrence of a in U . The third follows because two successive ω -blocks cannot both have cofinal image in the single target block. Also $U \not\preceq AB$: its tail would have to lie in either the pure a block or the pure b block, losing one of its recurrent labels. These facts show why both order of occurrence and cofinal type matter.

8.4 Negative controls: three invalid shortcuts

The hypotheses of Theorems 5.2 and 6.3 are not decorative. Each of the following three simplifications is tempting, and each fails on the two-letter antichain $P = \{a, b\}$ with an explicit witness. The three witnesses are also executed as negative controls in the verification suite of Section 22, where they are required to *fail*, so a regression that silently weakened a hypothesis would be caught.

Shortcut 1: omitting the guards. For any letter a ,

$$aB_P \simeq B_P, \quad \text{while} \quad a \not\preceq \varepsilon. \tag{8.1}$$

A component placed directly before a separator can therefore disappear under embeddability, so an unguarded tuple map need not reflect the componentwise order. This is (6.1) read as a counterexample.

Shortcut 2: using a finite guard at the universal block. Appending one letter does not repair the previous failure:

$$aaB_P \preccurlyeq aB_P, \quad \text{while} \quad a \not\preccurlyeq \varepsilon. \quad (8.2)$$

More generally, any fixed finite padding is absorbed into a universal periodic tail by (6.1). The guard of Lemma 6.1 is of limit length for a mathematical reason, not for convenience.

Shortcut 3: inserting a downset after one that contains it. The newness condition (4.1) cannot be dropped. Let the old family be $\mathcal{G} = \{E\}$ with $E = P$, and let the new downset be $D = \{a\}$, which is contained in E ; take the naive terminal guard $b \notin D$. Consider the two component pairs

$$(\varepsilon, a) \quad \text{and} \quad (B_E, \varepsilon).$$

The first is *not* componentwise below the second, since $a \not\preccurlyeq \varepsilon$. Nevertheless their guarded encodings satisfy

$$b a^\omega a b \preccurlyeq B_E b a^\omega b. \quad (8.3)$$

Indeed, match the initial source b inside B_E , then the source a^ω cofinally inside that same B_E ; then match the remaining source a into the displayed target a^ω and the final source b to the terminal target b . The source separator has used the *old containing* block rather than its designated target separator, and synchronization fails. Introducing downsets in nondecreasing cardinality, as in the proof of Theorem 7.1, is exactly what rules this out: a strictly earlier downset of no larger size cannot contain the new one.

The article asserts the necessity of $D \not\subseteq E$ in prose several times; (8.3) is the worked instance behind those assertions.

9 Consequences and finite-poset examples

9.1 Chains, antichains, and extremal values

Corollary 9.1. *For $n \geq 2$,*

$$o(s_{\omega^2}^F(C_n)) = \omega^{\omega^{2n-1}}, \quad o(s_{\omega^2}^F(A_n)) = \omega^{\omega^{n+2n-2}}.$$

Among n -element posets, the first value is the minimum, attained exactly by chains, and the second is the maximum, attained exactly by antichains.

Proof. A chain has exactly $n + 1$ downsets, its initial segments. Every subset of an antichain is a downset, so it has 2^n downsets. For any finite poset, the initial segments of a linear extension give $n + 1$ distinct downsets, and there are at most 2^n subsets in total.

For the equality statement at the lower end, fix a linear extension. If there are only $n + 1$ downsets, every downset is one of its initial segments. In particular all principal downsets are comparable by inclusion. Since $\downarrow x \subseteq \downarrow y$ implies $x \leq_P y$, every pair of elements is comparable, and P is a chain. At the upper end, if every subset is a downset, a strict relation $x <_P y$ would make $\{y\}$ fail to be a downset. Thus P is an antichain. The function of $j(P)$ in (1.1) is strictly increasing, so the ordinal extremal assertions follow. $\square$

9.2 Equivalent counting invariants

Every downset of a finite poset is generated by its maximal elements, which form an antichain. Conversely, the maximal elements of the downset generated by an antichain are that antichain. These maps give a bijection, including the empty set. Thus $j(P)$ may equally be described as the number of antichains of P .

Complementation gives a bijection between downsets and upsets. Passing to the dual poset exchanges downsets with upsets, so $j(P^{\text{op}}) = j(P)$. The formula therefore yields

$$o(s_{\omega^2}^F(P^{\text{op}})) = o(s_{\omega^2}^F(P)).$$

This is equality of maximal order types, not a claim that the word quasi-orders are isomorphic.

For a disjoint union $P \amalg Q$, choosing a downset is equivalent to choosing one downset in each component, hence $j(P \amalg Q) = j(P)j(Q)$. For an ordered sum $P + Q$, in which every element of P lies below every element of Q , one has $j(P + Q) = j(P) + j(Q) - 1$. The two possibilities are a downset contained in P , or all of P together with a nonempty downset of Q . These identities make (1.1) explicit for many recursively assembled alphabets.

9.3 A table of examples

In Table 1, $\rho(P) = n + j(P) - 2$ is the finite exponent parameter in $\omega^{\omega^{\rho(P)}}$.

Alphabet P	n	$j(P)$	$\rho(P)$	Maximal order type
Two-element chain	2	3	3	ω^{ω^3}
Two-element antichain	2	4	4	ω^{ω^4}
Three-element chain	3	4	5	ω^{ω^5}
Three-element V or its dual	3	5	6	ω^{ω^6}
Two-element chain plus an isolated point	3	6	7	ω^{ω^7}
Three-element antichain	3	8	9	ω^{ω^9}
Four-element chain	4	5	7	ω^{ω^7}
Four-element diamond	4	6	8	ω^{ω^8}
Four-element antichain	4	16	18	$\omega^{\omega^{18}}$
Five-element antichain	5	32	35	$\omega^{\omega^{35}}$

Table 1: Exact consequences of Theorem 1.1. The diamond has one bottom element, two incomparable middle elements, and one top element.

The alphabet cardinality by itself is not enough: the two binary values already differ. Conversely, nonisomorphic alphabets can have the same maximal order type. For instance the three-element V and its dual have the same n and j , while a four-element chain and a three-element chain plus an isolated point have the same value of $n + j - 2$ despite different cardinalities.

9.4 Finite quasi-orders

For a finite quasi-order rather than a poset, first pass to its quotient by mutual comparison. Projection of labels preserves and reflects word embedding: a comparison of quotient labels is exactly the comparison of any representatives. Therefore the theorem applies with n equal to the number of equivalence classes and j equal to the number of downsets of the quotient poset. Counting raw labels instead of classes can give an incorrect answer.

10 Finite-height ordinal accounting

The common word, quotient and maximal-order-type conventions are those of Section 2. We use two additional explicit facts:

$$u \preceq v \implies |u| \leq |v|, \quad u \equiv v \implies |u| = |v|. \quad (10.1)$$

An increasing injection between ordinals implies the first inequality; apply it in both directions for the second. In addition to the product and disjoint-union identities in Theorem 2.2, the classical residual calculus gives

$$o(X) = \sup_{x \in X} (o(X \setminus \uparrow x) + 1). \quad (10.2)$$

Here an order-reflecting map into a wqo gives a lower bound for the target maximal order type. The finite-alphabet Higman formula used in both routes is Equation (2.1); a derivation from the classical residual calculus is supplied in Section E.

10.1 The ordinal amplification identity

The numerical calculation used repeatedly below is simple but important. For $m, d \geq 1$,

$$H_m^{\otimes d} = \omega^{\omega^{m-1} \cdot d}. \quad (10.3)$$

Indeed, multiplying monomials naturally adds their exponents naturally, and the natural sum of d copies of ω^{m-1} is $\omega^{m-1} \cdot d$. The countable *ordinary* ordinal sum satisfies

$$\sum_{d=1}^{\infty} H_m^{\otimes d} = \omega^{\omega^{m-1} \cdot \omega} = \omega^{\omega^m} = H_{m+1}. \quad (10.4)$$

The same expression is obtained by indexing $d = m' + 1$ for $m' \geq 0$. No countable natural product is being invoked.

Lemma 10.1 (Layered products). *Suppose a wqo Y contains induced copies Y_d of E^d for all $d \geq 1$, where $o(E) = H_m$. Suppose the copies are pairwise disjoint on the quotient and*

$$x \in Y_d, y \in Y_e, x \leq y \implies d \leq e.$$

Then $o(Y) \geq H_{m+1}$.

Proof. For each d , choose a maximal linear extension of Y_d . Order the layers by increasing d and use the chosen extension within each layer. The hypothesis guarantees that this is a linear extension of their *induced union*. Its order type is the left side of (10.4), by the product rule. The induced union is a suborder of Y , so its type gives a lower bound for $o(Y)$. There is no assertion that this particular linear extension must extend to all of Y with the same type. $\square$

11 Canonical indecomposables at finite height

This section supplies a self-contained bounded-height normal form. It is included to make the integer $N_k(P)$ effective and to identify precisely which generators are used by the lower-bound proof. The general tree/sequence correspondence is established in [5].

11.1 Indecomposability and finite concatenations

For a word $u: \alpha \rightarrow P$ and $\xi < \alpha$, its tail at ξ is the restriction to $[\xi, \alpha)$, reindexed by that interval's order type.

Definition 11.1. A nonempty word u is indecomposable if it is equivalent to every one of its nonempty tails.

If $|u| < \omega^k$ and u is indecomposable, then $|u| = \omega^r$ for some $r < k$. To see this, write $|u|$ in Cantor normal form. Unless it is a single power of ω with coefficient one, it has a nonempty tail of smaller order type. That contradicts (10.1). The property is invariant under equivalence: an embedding between two words of the same indecomposable ordinal length is cofinal, so a tail in one word receives a tail of the other.

Lemma 11.2 (An indecomposable lands in one factor). *If z is indecomposable and*

$$z \preceq u_1 u_2 \cdots u_t$$

for a finite list of words, then $z \preceq u_i$ for some i .

Proof. Among the finitely many target factors met by the embedding, choose the last one. From any source position whose image lies in that factor onward, all source positions map into the same factor. Thus a nonempty tail of z embeds in it. That tail is equivalent to z . $\square$

The lemma does not say that the original embedding places the entire source inside one factor. It says that *some* embedding does so. This distinction will also matter for separators.

11.2 Periodic terms and their order

Start with a leaf p of height zero for each $p \in P$. Given a finite nonempty set S of previously constructed terms, introduce

$$\Omega(S), \quad \text{ht}(\Omega(S)) = 1 + \max_{s \in S} \text{ht}(s).$$

We will retain only sets S that are antichains under the already defined term order. Choose any enumeration $S = \{s_1, \dots, s_t\}$ and define

$$\text{ev}(p) = p, \quad \text{ev}(\Omega(S)) = (\text{ev}(s_1) \cdots \text{ev}(s_t))^\omega. \quad (11.1)$$

The particular enumeration changes neither the embedding class nor the comparison rule below. Every evaluation is indecomposable, and

$$|\text{ev}(s)| = \omega^{\text{ht}(s)}. \quad (11.2)$$

For a node, every tail contains all sufficiently late complete periods, which proves indecomposability. The length calculation follows by taking the largest height among the finitely many factors of a period and then multiplying the period length by ω on the right.

Proposition 11.3 (Comparison of periodic terms). *The following recursive rules agree exactly with semantic embedding:*

$$p \leq q \iff p \leq_P q, \quad (11.3)$$

$$p \leq \Omega(T) \iff (\exists t \in T) p \leq t, \quad (11.4)$$

$$\Omega(S) \leq p \text{ never}, \quad (11.5)$$

$$\Omega(S) \leq \Omega(T) \iff (\forall s \in S)(\exists t \in T) s \leq t. \quad (11.6)$$

In particular, comparison terminates by descending to lower-height terms.

Proof. The leaf rules follow directly from the presence or absence of a suitable letter. A positive-height word cannot embed into a word of length one.

For necessity in (11.6), fix one occurrence of a source factor $\text{ev}(s)$ inside a period of $\text{ev}(\Omega(S))$. There is a later source position, so the image of this factor is bounded in the target word. A bounded initial segment of a countable repetition of a finite period meets only finitely many complete or partial periods. Therefore $\text{ev}(s)$ embeds into a finite concatenation of target factors $\text{ev}(t)$. By lemma 11.2, it embeds into one of them. This holds for each $s \in S$.

Conversely, choose for each s a suitable t with $s \leq t$. Place successive source factors into suitably chosen target factors in successive fresh target periods. Repeating this construction through ω source periods gives an embedding. The argument with $S = T$ but different enumerations proves independence of the enumeration. $\square$

For a finite set S of terms, remove every element strictly below another element of S , retaining the maximal elements. This changes neither side of the Hoare comparison in (11.6). Thus every node has an antichain-reduced form. Moreover, if two finite antichains dominate one another, then they are equal: for $s \in S$, choose $t \in T$ and $s' \in S$ with $s \leq t \leq s'$; the antichain property forces $s = s'$, and antisymmetry forces $s = t$.

Induction on height now proves that the reduced terms form a genuine poset, with no distinct but equivalent terms. Leaves and nodes cannot be equivalent because their lengths differ.

11.3 Completeness and finiteness

Theorem 11.4 (Finite decomposition and canonical atoms). *For every finite P and integer $k \geq 1$:*

1. *there are finitely many reduced terms of height less than k ;*
2. *every P -word of length less than ω^k is equivalent to a finite concatenation of evaluations of these terms;*
3. *every indecomposable word of length ω^r , $r < k$, is equivalent to exactly one reduced term of height r .*

Consequently those terms, with the order in proposition 11.3, are precisely $I_k(P)$.

Proof. For $k = 1$ the words are finite and the only indecomposables are letters. Assume the result through height $r - 1$ and consider a word u of length ω^r , where $r \geq 1$.

Partition its domain into ω consecutive intervals of length ω^{r-1} . By induction, each resulting block is equivalent to a finite nonempty list of lower-height atoms. Replacing all blocks by those lists and flattening yields an ω -indexed list of atoms of heights at most $r - 1$. There are only finitely many available atom types.

Let S be the nonempty set of types occurring infinitely often in that list. Delete a finite initial part containing all occurrences of every type outside S . The remaining list is equivalent to a cyclic repetition of S : map each of its successive entries into a fresh target period in one direction, and choose successive sufficiently late occurrences of the required entries in the other. Replacing S by its maximal elements gives a reduced node $\Omega(S)$.

Every original block has length ω^{r-1} , so its finite factorization must contain an atom of height $r - 1$. Finitely many types are available; hence some such type recurs infinitely often. Therefore the reduced set S still has maximal height $r - 1$, and its node has height r . We have shown that u is equivalent to a finite lower-height prefix followed by one atom of height r .

If u itself is indecomposable, it is equivalent to that last atom. One way to justify this without assuming a particular factorization of u is to use [lemma 11.2](#): the finite prefix factors have length less than ω^r , so u cannot embed into them. It must embed into the final atom, which already embeds into u .

For an arbitrary ordinal $\alpha < \omega^{r+1}$, split α into finitely many blocks of length ω^r and one remainder of smaller length. The preceding argument handles the blocks and induction handles the remainder. This proves finite decomposition at the next level. There are only finitely many nonempty antichains of the finite poset of lower-height terms, so there are finitely many new terms. Uniqueness and exact comparison follow from [proposition 11.3](#) and antichain reduction. $\square$

11.4 The exact atom-count recurrence

Every order ideal of a finite poset is determined by its antichain of maximal elements, and every antichain determines an ideal. Empty objects correspond to one another. Thus $J(R)$ also counts the antichains of R .

All nonleaf terms of height at most k are in bijection with the nonempty antichains of $\mathcal{I}_k(P)$. This includes lower-height nodes already present: a node with lower-height children is the same node, not a new copy. Adding the n leaves gives

$$N_{k+1}(P) = n + J(\mathcal{I}_k(P)) - 1. \quad (11.7)$$

Equivalently, the number of *new* height- k atoms is $J(\mathcal{I}_k(P)) - J(\mathcal{I}_{k-1}(P))$. This formulation avoids double-counting old nodes.

Corollary 11.5 (The finite-generator upper bound). *If P is nonempty, then*

$$o(\mathcal{V}_k(P)) \leq H_{N_k(P)}.$$

More generally, if B is any finite nonempty set of distinct indecomposable word classes and $\mathcal{L}(B)$ is its finite-concatenation language, then $o(\mathcal{L}(B)) \leq H_{|B|}$.

Proof. Concatenation gives a monotone map from B^* onto $\mathcal{L}(B)$ up to equivalence. Apply the Higman formula and the monotone-surjection rule. For $B = \mathcal{I}_k(P)$, [theorem 11.4](#) identifies the image with $\mathcal{V}_k(P)$. The same argument proves that all these languages are wqo. $\square$

12 Cofinal blocks and protected final segments

Fix throughout this section

$$\lambda = \omega^r, \quad 1 \leq r < \omega.$$

Every nonempty final interval of λ has order type λ . Every bounded initial interval of λ has order type less than λ . It follows that an increasing embedding of a word of length λ into another word of that length has cofinal image.

12.1 The equal-block principle

Lemma 12.1 (Equal blocks force coordinate preservation). *Let $u_1, \dots, u_d$ and $v_1, \dots, v_d$ be words, all of length exactly λ . Then*

$$u_1 \cdots u_d \preceq v_1 \cdots v_d \iff u_i \preceq v_i \text{ for every } i. \quad (12.1)$$

More generally, an embedding from d such blocks into e such blocks requires $d \leq e$.

Proof. For each source block, consider the last target block it meets. A tail of the source block lies entirely in that target block. Its length is still λ , so its image is cofinal there. Successive source blocks must therefore have strictly increasing last target blocks: a later source position cannot be placed after a cofinal image while remaining in the same target block. This proves $d \leq e$.

When $d = e$, the strictly increasing assignment from source block indices to their last target block indices must be the identity. The first source block is contained in the first target block. After its cofinal image, no position of the second source block can lie in the first target block; and its last target block is the second. Continue inductively. Thus each source block lies wholly in its corresponding target block, proving necessity in (12.1). Sufficiency is componentwise concatenation of embeddings. $\square$

Indecomposability of the individual words is not assumed in this lemma. Their common ordinal length supplies the necessary cofinality. Equal numbers of blocks, however, are essential to its coordinate conclusion.

12.2 The protection condition

Definition 12.2. A word is λ -ended if its length is λd for some positive finite integer d .

Every nonempty tail of a λ -ended word has order type at least λ . More precisely, its length is λe for some positive finite e . A tail may contain several remaining λ -blocks; it need not have length exactly λ .

Lemma 12.3 (No bounded spill into a protected block). *Let u be λ -ended, and let z have length λ . In an embedding, a nonempty tail of u cannot be mapped into a bounded initial interval of a target occurrence of z .*

Proof. Such a tail has length at least λ , whereas the target interval has length less than λ . This violates (10.1). $\square$

This tiny observation is the protection mechanism. Without it, an old finite suffix can disappear into the beginning of a new infinite separator, destroying coordinate reflection.

13 Activating the first atom at a new level

The first atom of a new height needs a different construction from the later atoms. At this point there is not yet a large family of λ -ended words to place between separators.

Choose a minimal letter $a \in P$ and another letter $b \neq a$. Then $b \not\preceq_P a$. Put

$$c = a^\lambda.$$

This is an indecomposable of length λ . It is minimal among the indecomposables of that length: any word of length λ embedding into it can use only letters at most a , hence only the letter a .

Lemma 13.1 (A terminal marker protects the whole lower-level word). *The map*

$$F: \mathcal{V}_r(P) \longrightarrow \{\text{words of length } \lambda\}, \quad F(u) = u b a^\lambda \quad (13.1)$$

is an order embedding on equivalence classes. Its image is contained in $\mathcal{L}(\mathcal{I}_r(P) \cup \{c\})$.

Proof. Since $|u| < \lambda$ and λ is an infinite additively indecomposable ordinal, $|u| + 1 + \lambda = \lambda$. Concatenation proves monotonicity.

Suppose $uba^\lambda \preceq vba^\lambda$. The distinguished source occurrence of b cannot map into the target a^λ tail, because $b \not\preceq_P a$. It must map either to the distinguished target b or into v . Every position of u maps strictly before that position, and therefore inside v . Hence $u \preceq v$.

Finally, u is a finite concatenation of lower-height atoms up to equivalence, b itself is a lower-height atom, and $a^\lambda = c$. $\square$

It is not necessary that the distinguished source b map to the distinguished target b . The proof deliberately allows it to land earlier inside v ; either possibility recovers $u \preceq v$.

Proposition 13.2 (First-atom amplification). *Suppose $o(\mathcal{V}_r(P)) = H_N$, where $N = N_r(P)$. Then the language $\mathcal{L}(\mathcal{I}_r(P) \cup \{a^\lambda\})$ contains a λ -ended induced suborder of maximal order type H_{N+1} . The language itself has the same maximal order type.*

Proof. Let D be the image of F . By lemma 13.1, $o(D) = H_N$ and every word in D has length λ . For every $d \geq 1$, concatenate d independently chosen elements of D . By lemma 12.1, the resulting induced suborder is a copy of D^d .

The images for different d have distinct lengths λd , so their quotient classes are disjoint. Length comparison prohibits an embedding from a higher- d layer to a lower- d layer. Their union is λ -ended, and lemma 10.1 gives type at least H_{N+1} . There are exactly $N + 1$ available indecomposable generators, so corollary 11.5 gives the matching upper bound for the whole language and hence for this induced suborder. $\square$

This is the only point in the main induction where the availability of a second inequivalent letter is required. For a singleton alphabet no such b exists, and the much smaller order type in corollary 1.4 is the actual answer.

14 Separator amplification

We now prove the main lower-bound lemma. It permits new generators that dominate old ones, so it is not merely a construction with pairwise incomparable separators.

Lemma 14.1 (Protected separators). *Let B be a finite set of indecomposable word classes, each of length at most λ . Let z be an indecomposable word of length λ such that*

$$z \not\preceq b \quad \text{for every } b \in B. \quad (14.1)$$

Let $E \subseteq \mathcal{L}(B)$ be an induced suborder consisting of λ -ended words. For $m \geq 0$, define

$$\Phi_m(u_0, \dots, u_m) = u_0 z u_1 z \cdots z u_m, \quad u_i \in E. \quad (14.2)$$

Then Φ_m is an order embedding of E^{m+1} into $\mathcal{L}(B \cup \{z\})$. Moreover,

$$\Phi_m(\vec{u}) \preceq \Phi_t(\vec{v}) \implies m \leq t. \quad (14.3)$$

All outputs are λ -ended.

Proof. First, z does not embed into any word of $\mathcal{L}(B)$. Indeed, otherwise lemma 11.2 would embed it into one generator, contrary to (14.1).

Step 1: each source separator has a target separator as its last component. Consider an embedding of an output of Φ_m into an output of Φ_t . Regard the latter as a finite concatenation of its displayed old words v_j and displayed new separators z . Fix one displayed source separator and take the last displayed target component it meets. A tail of the source separator embeds into this component. Since the separator is indecomposable, the whole separator embeds into that component. The component cannot be an old word v_j . It must be a displayed target separator.

Step 2: the matching of separators is strictly increasing. The tail from Step 1 has length λ , so its image in that target separator is cofinal. No position of a subsequent source separator can follow that cofinal

image while remaining in the same target separator. Thus the last target separators of consecutive source separators are strictly increasing. There are t target separators, proving (14.3).

Step 3: with equal counts, the old coordinates cannot spill. Suppose $m = t$. The increasing assignment between two m -element index sets is the identity. Therefore the i th source separator ends cofinally in the i th target separator, for every i .

Consider an old source word u_i . If $i > 0$, the cofinal image of the preceding source separator prevents every position of u_i from landing in or before the preceding target separator. If $i < m$, all positions of u_i precede the image of the first position of the next source separator. That first image lies in or before the corresponding next target separator, because that separator is its last target component.

If any position of u_i entered the next target separator, all subsequent positions of u_i would also lie there, strictly below the image of the first position of the next source separator. This would embed a nonempty tail of u_i into a bounded initial interval of a length- λ block. It is impossible by lemma 12.3. Therefore u_i remains wholly inside v_i . The same argument handles u_0 without a left boundary; u_m has no right separator, but the previous separator already forces it into the final target component v_m .

Hence $\Phi_m(\vec{u}) \preceq \Phi_m(\vec{v})$ implies $u_i \preceq v_i$ for all i . Conversely, coordinatewise embeddings together with identity embeddings on z concatenate to an embedding of the outputs. This proves order reflection and preservation.

Finally, finite concatenation of words whose lengths are positive finite multiples of λ , interleaved with words of length λ , again has length a positive finite multiple of λ . $\square$

Remark 14.2 (Why the protection assumption is indispensable). Take the old language to be finite words and $z = a^\omega$. The source coordinate a does not embed into the empty target coordinate, yet $az \equiv z$. Thus merely requiring z not to embed into an old word is insufficient for coordinate reflection. It controls how many separators there can be, but not absorption before a separator. The λ -ended condition is exactly what rules out this failure.

Proposition 14.3 (One additional atom, one additional exponent step). *Under the hypotheses of lemma 14.1, suppose $|B| = M$ and $o(E) = H_M$. Then $\mathcal{L}(B \cup \{z\})$ contains a λ -ended induced suborder of maximal order type H_{M+1} , and its own maximal order type is H_{M+1} .*

Proof. Take the union of the images of Φ_m for all $m \geq 0$. By lemma 14.1, its layers are copies of E^{m+1} ; an embedding cannot decrease m . Different layers cannot be equivalent, since applying the separator-count inequality in both directions would force their indices equal. Lemma 10.1 therefore gives lower bound H_{M+1} .

Condition (14.1) ensures that z is not equivalent to a generator in B , so the enlarged generator alphabet has $M + 1$ elements. Apply corollary 11.5 for the matching upper bound. The union is λ -ended by the last assertion of lemma 14.1. $\square$

15 Proof of finite-alphabet exactness

We now assemble the argument for theorem 1.3. The finiteness, canonical order, and recurrence were proved in section 11; only equality in the order-type bound remains.

Proof of theorem 1.3. Induct on k . For $k = 1$, $\mathcal{V}_1(P) = P^*$ and $N_1(P) = n$, so the result is the finite-alphabet Higman formula.

Assume

$$o(\mathcal{V}_r(P)) = H_{N_r(P)}$$

for some $r \geq 1$, and set $\lambda = \omega^r$ and $N = N_r(P)$. Let Z be the finite poset of the new height- r atoms:

$$Z = \mathcal{I}_{r+1}(P) \setminus \mathcal{I}_r(P).$$

Choose a minimal letter $a \in P$. The constant word a^λ is a minimal element of Z , as shown in section 13. Choose a linear extension

$$z_1, z_2, \dots, z_d$$

of Z that starts with $z_1 = a^\lambda$. Define

$$B_i = \mathcal{I}_r(P) \cup \{z_1, \dots, z_i\} \quad (1 \leq i \leq d).$$

By [proposition 13.2](#), $\mathcal{L}(B_1)$ contains a λ -ended induced suborder E_1 of type H_{N+1} . Suppose, for $i \geq 2$, that $\mathcal{L}(B_{i-1})$ contains a λ -ended induced suborder E_{i-1} of type H_{N+i-1} .

The atom z_i cannot embed into a lower-height atom: its ordinal length is larger. It cannot embed into any z_j with $j < i$ either, because $z_1, \dots, z_d$ is a linear extension of the atom order. Hence $z_i \not\leq b$ for every $b \in B_{i-1}$. Apply [proposition 14.3](#) to obtain a λ -ended induced suborder $E_i \subseteq \mathcal{L}(B_i)$ of type H_{N+i} .

After all d new atoms have been added,

$$B_d = \mathcal{I}_{r+1}(P), \quad N + d = N_{r+1}(P).$$

Its concatenation language is all of $\mathcal{V}_{r+1}(P)$ up to equivalence, by [theorem 11.4](#). The constructed lower bound is $H_{N_{r+1}(P)}$, and [corollary 11.5](#) gives exactly the same upper bound. This completes the induction. $\square$

15.1 The stronger induction invariant

At every positive height the proof establishes slightly more than the statement of the theorem. Once the first atom of that height is activated, each intermediate language already realizes its full Higman upper bound on words ending in an entire block of that height. This invariant is what makes the next separator admissible without adding a new letter to P .

The proof does *not* exhibit an order embedding of the entire free word order $\mathcal{I}_k(P)^*$ into $\mathcal{V}_k(P)$. It exhibits a sequence of induced product layers large enough to attain the same maximal order type. Equality of maximal order types is weaker than an isomorphism of these orders, and absorption shows why that distinction is necessary.

16 Binary cases through the height-uniform construction

16.1 Two incomparable letters

Let $P = A_2 = \{a, b\}$. At heights zero and one the atoms are

$$a, \quad b, \quad A = a^\omega, \quad B = b^\omega, \quad C = (ab)^\omega.$$

The nontrivial relations are generated by

$$a < A < C, \quad b < B < C.$$

The two middle atoms A, B are incomparable, as are the two letters. There are exactly five distinct atoms, so the upper bound is H_5 .

The lower-bound construction can be read as three concrete stages. Start with finite binary words, whose type is $H_2 = \omega^\omega$. The map

$$u \mapsto uba^\omega$$

embeds them into words of length ω . Finite products of these blocks, layered by the number of blocks, give a family E_A of ω -ended words with type

$$\sum_{d \geq 1} (\omega^\omega)^{\otimes d} = \omega^{\omega^2} = H_3.$$

Only a, b, A are needed as generators.

Next add B . It embeds into neither a finite letter nor A , so it does not embed into any old word. Interleave m copies of B among $m + 1$ independently chosen members of E_A . The protected-separator lemma gives a new ω -ended family $E_{A,B}$ of type

$$\sum_{m \geq 0} H_3^{\otimes(m+1)} = H_4 = \omega^{\omega^3}.$$

Finally add C . It does not embed into any of a, b, A, B : an infinite repetition containing both letters cannot embed into a monochromatic infinite block. The same construction yields

$$\sum_{m \geq 0} H_4^{\otimes(m+1)} = H_5 = \omega^{\omega^4}.$$

This matches the five-generator upper bound and proves the binary antichain instance of [theorem 1.1](#).

Notice that C *dominates* the old infinite atoms. This does not invalidate its use as a separator. What matters is that C does not embed *into* an old atom; protection prevents the old coordinates from being swallowed by a bounded part of a target C .

16.2 Two comparable letters

For $C_2 = \{0 < 1\}$, the height-zero and height-one atoms are

$$0, \quad 1, \quad A = 0^\omega, \quad B = 1^\omega.$$

Here $(01)^\omega \equiv 1^\omega$, because every source letter is at most 1 and the 1 positions occur cofinally. The atom order is a diamond: 0 is below the incomparable atoms 1, A , both of which are below B .

The same activation map $u \mapsto u10^\omega$ gives type H_3 after A is added. Adding B as a protected separator gives H_4 . There are only four atoms, so

$$o(\mathcal{V}_2(C_2)) = \omega^{\omega^3}.$$

Thus two alphabets with the same finite maximal order type 2 already have different exact transfinite-word types.

16.3 All finite alphabets at height two

The height-one atoms correspond to nonempty order ideals of P . An antichain $S \subseteq P$ represents the periodic word $(\prod_{s \in S} s)^\omega$; its class depends precisely on the ideal $\downarrow S$. Therefore

$$N_2(P) = n + J(P) - 1,$$

which proves [theorem 1.1](#) from the main theorem.

17 Effective enumeration and computed examples

17.1 Canonical generation as a finite algorithm

The recurrence (11.7) comes with a direct algorithm. Store each leaf by its label and each node by a sorted tuple of child identifiers. At a new height, enumerate the antichains of the already constructed poset; retain those whose largest child height is the preceding height; and create the corresponding nodes. Compute comparisons by (11.3)–(11.6), memoizing each ordered pair of identifiers.

This is a decision procedure for all bounded-height atom comparisons. It also gives an exact ordinal notation for the maximal order type: a finite integer $N_k(P)$ together with the fixed expression $\omega^{\omega^{N_k(P)-1}}$. No enumeration of actual infinite words is required.

The finite objects can nevertheless become large. With N existing nodes, there are at most N^2 ordered pairs to memoize and at most N^2 child-pair checks per pair, giving a simple $O(N^4)$ arithmetic-operation bound for the complete comparison matrix. Antichain enumeration can have exponentially many outputs. The code uses integer bit masks and explicit size caps; it makes no claim of a polynomial-time algorithm in k or in the original alphabet size.

17.2 Two independent ideal counters

The package counts $J(R)$ in two ways. Let M be the set of remaining vertices in an induced subposet.

The first method counts antichains as independent sets of the comparability graph. For a chosen $x \in M$, an antichain either omits x or contains x and omits every vertex comparable with x :

$$A(M) = A(M \setminus \{x\}) + A(M \setminus \{y \in M : y \leq x \text{ or } x \leq y\}). \quad (17.1)$$

The second method directly partitions ideals according to whether they contain x :

$$J(M) = J(M \setminus \uparrow_M x) + J(M \setminus \downarrow_M x). \quad (17.2)$$

In the first branch, excluding x forces exclusion of everything above it. In the second, the entire downset of x is fixed as present, and one chooses an ideal in the remainder. Both recursions have value one on the empty poset and are memoized. For small posets the code additionally checks all subsets directly against the definition of an ideal.

The two counters share the mathematically specified atom order, but they do not share an antichain-enumeration implementation. Their agreement is therefore a meaningful check of the numerical computations, not an independent proof of the atom-comparison theorem.

17.3 Counts and exact ordinal parameters

Write A_n for the n -element antichain and C_n for the n -element chain. The following are exact values of $N_k(P)$; a dash means not computed in this package.

Alphabet	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$
A_2	2	5	11	28	136	—
C_2	2	4	7	12	22	50
A_3	3	10	43	1962	—	—
C_3	3	6	12	28	—	—

Table 2: Canonical indecomposable counts. Each entry N gives $o(\mathcal{V}_k(P)) = \omega^{\omega^{N-1}}$.

For example, the theorem and the table give

$$\begin{aligned} o(\mathcal{V}_3(A_2)) &= \omega^{\omega^{10}}, & o(\mathcal{V}_4(A_2)) &= \omega^{\omega^{27}}, \\ o(\mathcal{V}_3(C_2)) &= \omega^{\omega^6}, & o(\mathcal{V}_4(A_3)) &= \omega^{\omega^{1961}}. \end{aligned}$$

The last reported value for each family is obtained from the two exact ideal counts at the preceding level. Its full next-level atom poset is not materialized. The CSV records this distinction explicitly.

The first binary counts can be verified without software. The five-element $I_2(A_2)$ has nine ideals not containing C : choose a prefix in each of the two disjoint two-element chains $a < A$ and $b < B$. There is one more ideal containing C , namely the whole poset. Thus $J(I_2(A_2)) = 10$ and $N_3(A_2) = 2 + 10 - 1 = 11$.

The six height-two atoms in this case can be represented by

$$\begin{aligned} &\Omega(\{A\}), \quad \Omega(\{B\}), \quad \Omega(\{C\}), \\ &\Omega(\{A, B\}), \quad \Omega(\{A, b\}), \quad \Omega(\{a, B\}). \end{aligned}$$

For the two-element chain, $I_2(C_2)$ is the four-element diamond and has six ideals, giving $N_3(C_2) = 2 + 6 - 1 = 7$.

18 Growth, interpretation, and limits of the result

18.1 A direct iterated-exponential lower bound

The exact formula also supplies a simple finite-alphabet growth result. For $n \geq 2$, define integers

$$w_0(n) = n, \quad w_{r+1}(n) = \left(\frac{w_r(n)}{\lfloor w_r(n)/2 \rfloor} \right).$$

Proposition 18.1. *The height- r atoms over A_n contain an antichain of size $w_r(n)$. Consequently, for $k \geq 2$,*

$$N_k(A_n) \geq n + 2^{w_{k-2}(n)} - 1, \quad o(\mathcal{V}_k(A_n)) \geq \omega^{\omega^{n+2^{w_{k-2}(n)}-2}}. \quad (18.1)$$

Proof. The n leaves give the height-zero antichain. Given an antichain D of height- r atoms, the nodes $\Omega(S)$ for nonempty $S \subseteq D$ are ordered exactly by subset inclusion: no distinct members of D compare. Choosing all subsets of size $\lfloor |D|/2 \rfloor$ produces an antichain of the claimed size at height $r + 1$. For the stated n , this size is positive and the subsets used are nonempty.

Also, all $2^{|D|} - 1$ nonempty subsets give distinct nodes at the next height. Use a height- $(k - 2)$ antichain and retain those nodes together with the n leaves. This proves the first inequality in (18.1); the second follows from [theorem 1.3](#). $\square$

The elementary bounds

$$\frac{2^m}{m+1} \leq \binom{m}{\lfloor m/2 \rfloor} \leq 2^m$$

follow because the largest binomial coefficient is at least the average of the $m + 1$ coefficients. Thus, for fixed k and growing n , the finite exponent parameter has the expected iterated-exponential growth. The upper estimate

$$N_{k+1}(P) \leq n + 2^{N_k(P)} - 1$$

follows from (11.7). These are statements about finite alphabets and growth in their size; no sharp asymptotic equivalent is claimed.

18.2 The quotient, not the literal set of words

There may be uncountably many literal words of length ω over a finite alphabet, but at each fixed finite level there are only countably many equivalence classes of all words. Indeed, every class has a finite code over the finite atom alphabet. The large ordinals in the theorem measure ordering complexity of this quotient, not the cardinality of the original set of functions.

The integer $N_k(P)$ should also not be mistaken for the number of all word classes. It counts only indecomposable classes. Their finite concatenations have infinitely many classes and supply the much larger maximal order type.

18.3 Why this does not compute the all-height union

Although

$$\mathcal{V}_{<\omega^\omega}(P) := \{u : |u| < \omega^\omega\} = \bigcup_{k \geq 1} \mathcal{V}_k(P),$$

one cannot infer its maximal order type merely by taking the supremum of the numbers in [theorem 1.3](#). Maximal order type is not generally continuous under increasing unions of induced suborders.

A simple example is the increasing union of the finite squares $\{0, \dots, m-1\}^2$ under componentwise order. Each has maximal order type m^2 , so the supremum of their types is ω . The union is $\omega \times \omega$, whose maximal order type is $\omega \otimes \omega = \omega^2$. The all-height problem requires its own argument; the broader tree correspondence and all-height results in [5] are separate from the finite-level proof here.

18.4 What remains outside the theorem

The proof uses finiteness in two different ways. The normal-form step needs only finitely many lower-height atom classes, so that recurrent types can be isolated after a finite prefix. The exactness step adds all new atoms in a finite linear extension and uses the finite-letter Higman formula. Neither argument directly provides an exact answer for every infinite wqo alphabet, arbitrary transfinite height, or arbitrary cutoff ordinal that is not a power ω^k .

Nor does a large maximal order type automatically specify an optimal length-function bound for finite controlled bad sequences. Such a bound also requires a norm and a control function. The ordinal theorem is an input to that separate analysis, not a substitute for it.

19 Verification, proof audit, and reproducibility

19.1 What was actually checked

The delivered height-suite record was produced under Python 3.13.5, using only the standard library. It is written to support Python 3.9 or later. The fixed random seed is 20260919. The recorded run reports:

Check	Count
All labelled posets on at most three elements	24
Canonical atom posets through $k = 3$ for those alphabets	72
Agreements of two independent ideal counters	87
Agreements with direct exhaustive subset counting	64
Marker-map comparison pairs	1,922
Equal-block comparison pairs	10,000
Separator comparison pairs	18,000
of which equal-separator coordinate checks	13,489
Singleton length-order comparison pairs	625
Absorption sanity checks	2
Exact count-table entries	19

Table 3: The completed reproducibility run. These are finite checks, not a formal verification of the transfinite proof.

The count “24” includes one empty poset, one one-element poset, three labelled two-element posets, and nineteen labelled three-element posets. The atom checks verify reflexivity, antisymmetry, transitivity, height monotonicity, and antichain reduction. The symbolic word checks operate on exact ultimately periodic ω -blocks; they do not replace ω by a large finite integer. [Section F](#) describes the embedding algorithm and its correctness argument.

These checks exercise the finite representations, detect arithmetic and implementation mistakes, and test many instances of the structural lemmas. They do not establish arbitrary-height statements by exhaustive search. The latter are supported by the written induction and cofinality proofs.

19.2 The main points audited in the proof

The first possible failure is assuming that a surjective concatenation map is order-reflecting. It need not be; the proof uses it only for the upper bound. Order reflection for the lower bound is proved separately on the protected families.

The second is confusing “has the same cardinality” with “has the same ordinal length.” All no-spill arguments compare ordinal order types. A tail of a λ -ended word has length at least λ , and cannot fit below a single position of a target λ -block.

The third is silently treating an indecomposable as wholly contained in its last target factor. The original embedding may straddle factors. [Lemma 11.2](#) extracts an equivalent tail to obtain a new embedding; [lemma 14.1](#) then uses cofinality of that tail in its last target separator.

The fourth is choosing separators in the wrong order. The requirement is that the *new* atom not embed into any *old* atom. An increasing linear extension of the same-height atom poset supplies exactly that direction. Old atoms are allowed to embed into the new one.

The fifth is combining product layers without checking cross-layer comparisons. In activation, length makes the layer index monotone. In separator amplification, [equation \(14.3\)](#) does so. This also precludes identifying outputs from two different layers on the quotient.

Finally, the singleton alphabet is checked separately. It has one atom at each height, so $N_k(1) = k$, but its word type is ω^k , not H_k for $k \geq 2$. The missing second letter is a genuine mathematical obstruction to the activation argument, not a cosmetic exception.

19.3 How to reproduce the package

From the archive root, run

```
python code/heights/verify.py --out data/heights
```

The program regenerates `data/heights/verification.json`, the CSV count table, and explicit JSON atom posets. To inspect a particular alphabet, for example the binary antichain through height two, run

```
python code/heights/atoms.py --kind antichain --size 2 --k 3 --output atoms.json
```

A custom finite poset may be supplied as JSON containing distinct labels and a full reflexive Boolean relation matrix. The default cap prevents materialization of more than 5,000 atoms; the cap can be changed explicitly. Counting can still be expensive before a large object is materialized.

Build this document using

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The bibliography is included in the source, so no bibliography database or network access is needed. A separate `data/heights/references.bib` is provided for reuse. The original third-party papers and font files are not included.

20 Exact-length strata

This section computes the maximal order type of each individual length stratum. The results are supplementary: they are not used anywhere in the proof of Theorem 1.1, and a reader checking only the resolution of the two-letter question may skip the section entirely. Their point is negative, and it is made in Section 20.3: the answer of Theorem 1.1 cannot be assembled stratum by stratum.

Unlike the main line, this section does use the natural-product and natural-sum identities of Theorem 2.2(iii) and (iv). The elementary substitute of Lemma 6.7 is a one-sided bound and does not suffice here.

Throughout, P is a nonempty finite poset with $n = |P|$, and

$$\mathcal{E}_\alpha(P) = \{u : \text{len}(u) = \alpha\}, \quad d = j(P) - 2$$

is the number of *proper* nonempty downsets of P . Thus $d \geq 1$ exactly when $n \geq 2$, and $d = 0$ for a singleton alphabet.

20.1 A single infinite block

Proposition 20.1 (Length exactly ω). *For every nonempty finite poset P ,*

$$o(\mathcal{E}_\omega(P)) = H_n \cdot d + 1, \tag{20.1}$$

$$o(s_{\omega+1}(P)) = H_n \cdot (d + 1) + 1, \tag{20.2}$$

where $s_{\omega+1}(P)$ denotes the words of length at most ω .

Proof. By Lemma 3.4, every word of length ω is equivalent to pB_D with $D = I(u)$ and p empty or ending outside D . Group these classes into *fibres* indexed by the nonempty downset D .

Fix a proper nonempty D and consider its fibre. For two normalized forms,

$$pB_D \preccurlyeq qB_D \iff p \preccurlyeq q. \tag{20.3}$$

The implication from right to left is monotonicity of concatenation. For the other direction, the claim is trivial when $p = \varepsilon$; otherwise the final letter of p lies outside D , so by Lemma 3.1 it cannot be matched inside the target periodic tail, and its image lies in the target prefix q . Every earlier position of p maps strictly before it, hence also into q .

By (20.3) the map $p \mapsto pB_D$ is an isomorphism from the suborder of P^* consisting of the admissible prefixes, that is the words that are empty or end outside D , onto the fibre of D . Hence that fibre has type at most $o(P^*) = H_n$. In the other direction choose $c \notin D$ and map $u \in P^*$ to ucB_D , whose prefix uc is admissible. This map is order preserving, and order reflecting by (20.3) together with Lemma 5.1. Hence the fibre of D has type exactly H_n .

The fibre of the full downset P is a single equivalence class: by (6.1) every pB_P is equivalent to B_P . That class lies above every word of length ω , since $D \subseteq P$ always holds. Conversely, by Lemma 4.1, an embedding between words of length ω forces inclusion of their recurrent downsets, so comparisons across fibres go only from smaller to larger downsets.

For the lower bound, order the d proper nonempty downsets by nondecreasing cardinality, concatenate maximal linear extensions of their fibres in that order, and place the single class B_P last. No comparison is violated, because a comparison between distinct fibres always runs from a downset of strictly smaller cardinality to a larger one. This is a linear extension of type $H_n \cdot d + 1$.

For the upper bound, erase every comparison between distinct fibres. The result is a disjoint union of d orders of type H_n and one singleton, whose maximal order type is $H_n \oplus \dots \oplus H_n \oplus 1 = H_n \cdot d + 1$ by Theorem 2.2(iv); the natural sum of d copies of $\omega^{\omega^{n-1}}$ with 1 collects the coefficients at one exponent and leaves the finite part alone. Deleting comparisons cannot decrease the maximal order type, so the original order has type at most this value. This proves (20.1).

For (20.2), adjoin the finite words. No infinite word embeds into a finite word, so placing all of P^* first respects every comparison and adds one further layer of type H_n ; erasing the comparisons between P^* and $\mathcal{E}_\omega(P)$ gives the matching upper bound, again by Theorem 2.2(iv). $\square$

20.2 Several blocks and a fixed finite tail

Lemma 20.2 (Rigidity at equal block count). *Let u and v each have exactly r canonical ω -blocks, and suppose $u \preceq v$. Then the i th canonical source block is mapped, entirely and cofinally, inside the i th canonical target block, for every $i < r$, and the finite source tail is mapped inside the finite target tail.*

Proof. By Lemma 4.1, applied to the periodic normal form of each canonical source block, each source block has a cofinal destination among the target blocks, distinct source blocks have distinct destinations, and the assignment is increasing. An increasing injection of an r -element chain into itself is the identity, so the i th source block is cofinal in the i th target block.

The i th source block cannot have an image point before the end of the $(i-1)$ st target block, because the $(i-1)$ st source block has already used that block cofinally and every point of the i th source block comes later. It cannot have an image point after the end of the i th target block either: if it did, the infinitely many source positions following that one would map beyond the i th target block, leaving only finitely many image points inside it, contradicting cofinality there. Hence the whole i th source block lies inside the i th target block. After the last block only the finite target tail remains, which handles the tail. $\square$

Lemma 20.2 is strictly stronger than Lemma 4.1 with Remark 4.2, and it is not available in the main argument: it needs the hypothesis that source and target have the same number of canonical blocks, which is exactly what the product constructions of Sections 5 and 6 do not assume. Within the present section that hypothesis is automatic, because the length is fixed.

Proposition 20.3 (Fixed-length product decomposition). *For all $r, k < \omega$ there is an isomorphism of quotient orders*

$$\mathcal{E}_{\omega r+k}(P) \cong \mathcal{E}_\omega(P)^r \times P^k, \quad (20.4)$$

and consequently, writing $A = H_n \cdot d + 1$,

$$o(\mathcal{E}_{\omega r+k}(P)) = A^{\otimes r} \otimes n^k. \quad (20.5)$$

Proof. Concatenation is a bijection between r -tuples of length- ω words together with a length- k tail and words of length $\omega r + k$, since the canonical block decomposition of such a word is unique. By Lemma 20.2, an embedding between two such words restricts to an embedding of the i th block into the i th block and of the tail into the tail; conversely, componentwise embeddings concatenate by Lemma 2.1. A strictly increasing injection between finite chains of the same length k is the identity, so the induced relation

on the tails is exactly the componentwise label order of P^k . This proves (20.4), and (20.5) follows from Theorem 2.2(iii), Proposition 20.1, and $o(P^k) = n^k$ for the finite poset P^k . $\square$

Because $H_n = \omega^{\omega^{n-1}}$, expanding the natural power in (20.5) gives the Cantor normal form

$$o(\mathcal{E}_{\omega r+k}(P)) = \sum_{i=r, r-1, \dots, 0} \omega^{\omega^{n-1} \cdot i} \left(n^k \binom{r}{i} d^i \right). \quad (20.6)$$

The sum is written in decreasing i , so it is an ordinary Cantor-normal-form sum, and terms with zero coefficient are omitted. The coefficient arises as follows: in the expansion of $(H_n \cdot d + 1)^{\otimes r}$ one chooses, from each of the r factors, either the leading monomial $H_n \cdot d$ or the constant 1; choosing the leading monomial in exactly i of the factors can be done in $\binom{r}{i}$ ways, contributes the finite coefficient d^i , and produces the exponent $\omega^{n-1} \oplus \dots \oplus \omega^{n-1}$ with i summands, namely $\omega^{n-1} \cdot i$. The convention $d^0 = 1$ is in force, including when $d = 0$: for a singleton alphabet (20.6) correctly collapses to $n^k = 1$.

20.3 A warning about taking suprema over lengths

Suppose $n \geq 2$, so that $d \geq 1$. Taking the supremum of (20.6) over all lengths gives

$$\sup_{r, k < \omega} o(\mathcal{E}_{\omega r+k}(P)) = \omega^{\omega^n}, \quad (20.7)$$

because the leading exponents are $\omega^{n-1} \cdot r$, whose supremum over r is ω^n , and finite coefficients and lower-order terms do not affect that supremum.

The value (20.7) is *strictly below* the answer $\omega^{\omega^{n+j(P)-2}}$ of Theorem 1.1. To see the strictness, take a linear extension of P : its $n+1$ prefixes are distinct downsets, so $j(P) \geq n+1$ and therefore $n+j(P)-2 \geq 2n-1 > n$ whenever $n \geq 2$.

There is no contradiction, and the explanation is the one already given in Section 7.1. The length strata are *not* ordinal-sum layers of the whole order. Ordinal length is indeed monotone under embeddability, by Lemma 4.1 for the block count and Lemma 20.2 for the tail. But an ordinal-sum layering would require every word of a shorter length to embed into every word of a longer one, and that fails at once: over the two-letter antichain, b has length 1 and a^ω has length ω , yet $b \not\leq a^\omega$. A maximal linear extension of the full order is therefore free to interleave strata and exploit incomparabilities between them, and it does. The downset induction of Section 7 detects that additional complexity; the separate exact-length calculations, by construction, cannot.

21 An exact symbolic embedding algorithm

21.1 Representation

The supplied module `code/omega2.py` represents a word by a finite sequence of tokens. A token $(\text{'f'}, x)$ is a finite letter, and $(\text{'w'}, D)$ is B_D , with D stored as a nonempty bitmask downset. The poset relation is supplied by predecessor masks. The code validates reflexivity, antisymmetry, and transitivity.

This representation does not claim to compute the recurrent set of an arbitrary black-box infinite input. Rather, it decides embeddings between already supplied finite periodic-block descriptions. The mathematical normal-form theorem says such descriptions represent every equivalence class in the target order.

21.2 Greedy decision procedure

Maintain a pointer to a target token, initially the first. Read source tokens from left to right.

Source token is a finite letter a .

Scan target tokens until finding either a finite letter b with $a \leq_P b$, or an infinite token B_E with $a \in E$. In the finite case consume the letter and advance the pointer. In the infinite case leave the pointer at B_E : after finitely many matches, an equivalent periodic tail remains.

Source token is an infinite block B_D .

Scan target tokens until finding an infinite token B_E with $D \subseteq E$. Consume its tail cofinally and advance the pointer to the following token.

If the required token does not exist, reject. Accept after every source token has been processed.

The same rules, written as a table of the four possible source/target token pairs, are the following. A letter token is written a or b and an infinite token is written $[D]$ or $[E]$, denoting B_D and B_E .

Source	Target	Matching condition	Cursor after a match
a	b	$a \leq_p b$	Advance past b
a	$[E]$	$a \in E$	Remain at $[E]$
$[D]$	$[E]$	$D \subseteq E$	Advance past $[E]$
$[D]$	b	Never	No match

When the current target token cannot match, skip it and advance the cursor. The last row is worth stating explicitly: an infinite source token never matches a finite target token, since a strictly increasing map out of ω cannot land in a single position. The second row is the only one that leaves the cursor where it is, because a finite match consumes only finitely many positions of a periodic tail and leaves an equivalent tail behind.

Theorem 21.1 (Exactness of the algorithm). *The greedy procedure accepts exactly when the represented source ordinal word embeds into the represented target ordinal word.*

Proof. Within a periodic target block, removing any finite initial segment leaves an equivalent tail. Thus the symbolic state need not record a phase or a finite offset in that block.

For a finite source letter, the first eligible target token provides an early possible match. Choosing a later token cannot leave a more useful suffix. Within an infinite token, choosing an actual sufficiently early eligible occurrence leaves an equivalent periodic tail. Therefore a successful continuation after any possible match can also be made after the greedy match.

For an infinite source block B_D , Lemma 4.1 shows that every embedding must use some target block B_E cofinally with $D \subseteq E$. All later source positions must then lie after that target block. The greedy choice takes the first eligible such block. The entire source B_D embeds in its available tail by Lemma 3.1; allowing part of the source block to occur earlier cannot improve the endpoint beyond this first eligible block. The remaining target suffix after the greedy choice contains the suffix left by any later choice.

Inducting on the number of remaining source tokens proves completeness: whenever a completion exists, the greedy step preserves the existence of a completion. Conversely, each accepted finite step can be realized by an actual letter match in a finite token or a periodic tail, and each accepted infinite step by a cofinal embedding into the selected target block. Concatenating these realizations gives an ordinal embedding, so acceptance is sound. $\square$

The target pointer never retreats. With bit operations and relation lookup counted as unit cost, the decision procedure uses $O(|u|_{\text{tok}} + |v|_{\text{tok}})$ time and constant auxiliary space. For unbounded alphabet size, bitmask operations have their ordinary bit costs; validating masks and the poset incurs additional cost. The reference checker uses memoized exploration of every eligible target endpoint rather than the first endpoint.

21.3 Canonical quotient representatives

Proposition 3.5 produces finite expressions but, without its normalization clause, not unique ones. The normalization is easy to carry out on token expressions, and it makes the representatives canonical.

The normalization procedure. Process the tokens from left to right onto a stack. A letter token is pushed. When an ideal token $[D]$ arrives, first pop the immediately preceding letter tokens for as long as their letters belong to D , and then push $[D]$. The final finite tail, that is the letters following the last ideal token, is never altered. Equivalently: in each finite prefix preceding an ideal token $[D]$, delete the longest suffix all of whose letters lie in D .

Each deleted suffix is absorbed by the ideal token that follows it, by (3.2), so the output is equivalent to the input; and the output is exactly an expression of the shape (3.4), since every prefix is now empty or ends outside the downset of the block that follows it. The procedure is linear in the number of tokens and is implemented as `normalize` in `code/omega2.py`.

Proposition 21.2 (Uniqueness of normalized expressions). *Two normalized token expressions over a finite poset P are mutually embeddable if and only if they are identical.*

Proof. One direction is trivial. For the other, suppose u and v are normalized and mutually embeddable. Ordinal length is monotone under embeddability, so u and v have equal lengths; hence they have the same number r of canonical ω -blocks and finite tails of the same length.

By Lemma 20.2, applied in both directions, the i th block of u and the i th block of v are mutually embeddable, and so are the two finite tails. Mutual embeddability of blocks forces mutual inclusion of their downsets by Lemma 3.1, so the i th ideal tokens agree, say both equal to $[D_i]$. Since both expressions are normalized, the finite prefixes p_i and q_i preceding these tokens are empty or end outside D_i , so (20.3) turns the mutual embeddability of the blocks into mutual embeddability of p_i and q_i .

Finally, two finite words over a poset that embed into each other have equal finite length, so both index maps are the identity; antisymmetry of $\leq_P$ then forces equality of corresponding labels. Hence $p_i = q_i$ for every i , the tails agree, and $u = v$. $\square$

Consequently the quotient of $s_{\omega^2}^F(P)$ has a countable, effectively comparable system of canonical finite representatives, even though there are uncountably many literal infinite words over an alphabet with at least two elements. The forward reference promised after Proposition 3.5 is discharged here.

21.4 Why finite expansion is not an adequate substitute

Replacing each B_D by M repetitions for a large integer M changes the problem. The absorption equivalence $aB_{\downarrow a} \simeq B_{\downarrow a}$ already fails for the direction from length $M + 1$ to length M after a uniform finite expansion. Conversely, finite approximations do not directly certify the cofinal-boundary restrictions needed for reflection of product maps. The supplied code therefore treats ω -blocks symbolically and uses the proven cofinal semantics exactly.

22 Computational checks and reproducibility

The included verification run passed 3,240,040 assertions. Its random seed is 20260919; all random checks supplement deterministic checks. The report is stored in `data/verification.json`. The recorded run used Python 3.14.4, and the code targets Python 3.10 or newer without third-party packages. The suite is a single program: the counts below come from one run, and they are not the sum of two separately reported runs.

The exhaustive comparison of the two deciders uses all naturally labelled posets of sizes zero through three and every source/target token word of length at most two. The longer-word comparisons cover all naturally labelled posets of sizes two through four and use token words of length at most twelve.

22.1 The exhaustive binary domain

Over the two-letter alphabets the whole token domain up to length four is enumerated and every ordered pair is tested. The antichain has five tokens, giving $1 + 5 + 5^2 + 5^3 + 5^4 = 781$ expressions and $781^2 = 609,961$ ordered pairs; the chain has four tokens, giving 341 expressions and 116,281 pairs, for 726,242 ordered pairs in all. For each pair the greedy and reference comparators agree, normalization preserves the decided relation, and mutually embeddable expressions have identical normal forms, which is the executable form of Proposition 21.2.

Across the same two alphabets, every allowed support subfamily is used as an insertion chain, giving 14 activation stages; the universal-only exceptional activation is excluded, as the hypotheses require. Support prefixes shared by different subfamilies are deliberately retested.

Check	Number of assertions
Exhaustive greedy/reference agreement, small posets	38,107
Exhaustive greedy/reference agreement, binary domain	726,242
Seeded longer-word greedy/reference agreement	9,800
Normalization preserves every decided relation	726,242
Normalization preserves the class and the ordinal length	2,244
Mutually embeddable pairs have identical normal forms	4,428
Exhaustive proper/universal product maps	1,544,960
Binary support-family product maps	44,667
Seeded longer-word product maps	35,500
Padding reflection	55,150
Padding has nonzero limit length	4,024
Universal-only comparison formula	20,640
Universal-only normalization	2,660
Fixed-length product decomposition	25,000
Maximal elements generate their downset	356
Boundary, absorption, and regression cases	16
Negative controls, required to fail	4
Total	3,240,040

Table 4: Executed symbolic checks. A “word length” in the test generator means token length, not ordinal length.

The three invalid shortcuts of Section 8.4 are executed as negative controls: the suite asserts that the naive encodings do *not* reflect the componentwise order, so weakening a hypothesis would be detected rather than silently accepted. The fixed-length decomposition of Proposition 20.3 is checked on randomly generated words of prescribed length $\omega r + k$ over every naturally labelled poset of size at most four.

For the product embeddings, every naturally labelled poset of sizes two through four is processed, introducing all nonempty downsets in cardinality order. There are 355 marker stages. At each stage the exhaustive test uses a fixed four-word component pool and all pairs of tuples of arities two and three. The seeded tests add tuples of arities two through five with component token lengths at most eight. In total these give 1,625,127 product-map comparisons. Separate padding tests allow all one-token words, including already universal words, so padding reflection is checked beyond the narrower family needed in the main induction.

The two embedding implementations share the mathematical periodic-tail semantics. Agreement between them is a check of the greedy optimization, not an independent derivation of that semantics. Likewise, no collection of the finite assertions in Table 4 proves ω^{ω^4} to be the maximal order type. That conclusion rests on the unbounded family of product embeddings and the ordinal calculation.

22.2 Enumeration conventions

The auxiliary enumerator considers relations on $\{0, \dots, n-1\}$ whose strict comparisons respect the usual numerical order. Every finite-poset isomorphism class has such a labelling, but the enumeration is neither an enumeration of all labellings nor an isomorphism-free enumeration. The numbers of generated relations for $n = 0, 1, 2, 3, 4, 5$ are respectively

$$1, 1, 2, 7, 40, 357.$$

These counts were computed by the included program. Histograms by the number of downsets appear in `data/ideal_counts.csv`. The 50 individual posets of sizes one through four, with their predecessor and successor masks, downset counts, and the exponent predicted by (1.1), are listed in `data/finite_posets.csv`; the row for $n = 1$ carries the literal entry `exception: omega^2` rather than a numerical exponent. Table examples are also exported to `data/examples.csv`. No claim about uniform sampling of isomorphism classes is intended.

To compute $j(P)$ for one small input poset, the code simply checks all 2^n subsets for downward closure. This exponential enumeration is sufficient for the examples and the proof tests. The article does not claim a polynomial-time algorithm for downset counting.

22.3 Reproduce the package

From the root of the unpacked directory:

```
python3 code/verify.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The Python command regenerates the verification reports and CSV tables. It accepts `-seed`, `-samples`, `-pair-bound`, and `-output-dir`; the defaults are 20260919, 250, 4, and `data/`. Raising `-pair-bound` enlarges the exhaustive binary enumeration very quickly. A three-pass `pdflatex` build is also supplied as `build.sh`. The PDF can alternatively be built by running `pdflatex article.tex` repeatedly until cross-references stabilize. No external bibliography processor is required; a separate BibTeX file is supplied for reuse.

23 Proof audit, scope, and further directions

23.1 Where a hidden gap would have to occur

The proof is organized so its nonclassical burden is concentrated in a few claims. The atom-count upper bound is a monotone-surjection argument. Cofinal localization uses finiteness of the target token decomposition. Synchronization uses the newness condition $D \not\subseteq E$ for all old blocks, not merely distinctness of block labels. The proper-marker reflection uses downward closure to exclude a guard from the next marker. The full-marker reflection uses *both* cancellation and the limit length of the padded component.

These dependencies explain several tempting but incorrect shortcuts. A source infinite block need not land wholly inside one target token. The ordinary evaluation map from atom strings need not reflect order. The universal marker admits no excluded finite letter. A finite test cannot certify an infinite supremum. None of these shortcuts is used.

The four boundaries to check independently. Concretely, four points carry the argument, and a hidden gap would have to sit at one of them.

- (1) Embeddings are *strictly* increasing on ordinal positions. Without strictness, cofinal consumption of a target block fails and Lemma 4.1 is false as stated.
- (2) A newly introduced downset is contained in *no* old downset, condition (4.1). Mere distinctness is not enough, and (8.3) is the explicit counterexample.
- (3) Exclusion applies to the *explicit pure periodic segment* of a displayed separator, not to the finite prefix that may precede it inside the same canonical block. This is Remark 4.3.
- (4) Full support requires an *earlier proper* support, so that a guard of positive limit length can be built. Without one, the exceptional value (7.2) is the truth.

The negative controls of Section 8.4 exhibit the failure of the second and fourth safeguards, and of guarding altogether; the singleton alphabet illustrates the resulting exceptional maximal order type.

A reader's guide. An independent referee can focus first on Theorems 5.2 and 6.3, or equivalently on their unified form Theorem 6.5. Once their product-reflection assertions are accepted, the classification follows by a short finite induction and the classical finite-product formula, or by the elementary Corollary 6.8 if one prefers to avoid that import. The binary specialization in Section 8 provides a second presentation of the same core argument, not an independent proof with different foundations.

Section 20 may be omitted entirely by a reader checking only the resolution of the original two-letter question. It is the one part of the article that uses the natural-product and natural-sum identities

Theorem 2.2(iii) and (iv) essentially, and nothing in Sections 1–9 depends on it. Its role is to document, in (20.7), why the answer cannot be obtained one length at a time.

23.2 Established within the argument versus not claimed

Within this manuscript, the main theorem has a complete proof relative to the classical results stated in Theorem 2.2. There is no remaining conjectural lemma inside that derivation. Nevertheless the argument has not undergone independent expert review or proof-assistant verification. The appropriate external status is therefore a *proposed solution*, with explicit hypotheses and reproducible supporting checks, rather than a certified resolution.

The article does not compute all invariants of these quasi-orders: their height, width, and exact isomorphism classification are different tasks. It does not show that two alphabets with equal $n + j(P)$ produce isomorphic word orders. It does not give a canonical representation for arbitrary infinite input streams. It also does not assert that selected block families at higher ordinal cutoffs have the same counting law.

23.3 Why the selected-family proof stops at this cutoff

The decisive compactness feature is that a target word below ω^2 has only finitely many ω -blocks. The image of a source ω -block must eventually stabilize in one of them. At length ω^2 , a source ω -block may instead visit infinitely many successive target ω -blocks, taking only finitely many points from each. Thus the cofinal localization lemma in its present form is no longer valid when that target length is allowed.

The full-language extension in Sections 12–15 supplies a hierarchy of equal-length blocks and protected separators. It does not assert the same classification for arbitrary selected families at higher cutoffs; that extension still needs a separate argument. Counting indecomposable blocks without that theorem would reproduce the same logical mistake that naive atom counting makes at the universal-only stage.

For infinite alphabets, even when every individual word has finite image, the set of possible recurrent downsets need not be finite, and a single finite inclusion ordering does not exhaust the construction. The present finite induction therefore does not directly transfer. These observations identify precise missing mechanisms for extensions; they are not reasons to reject further investigation.

23.4 A concrete formal-verification target

A useful formalization would first define ordinal concatenation and its embedding relation for finite periodic-token codes, then prove the exact symbolic decision theorem. The next targets are the cofinal boundary lemma, proper-marker reflection, and universal-marker reflection. With the classical maximal-order-type product and finite-word theorems available, the final induction has very little additional complexity. The package contains executable tests but no Lean development, and it makes no claim of machine-checked proof.

24 Conclusion

The finite generator upper bound for words below ω^2 is sharp for every finite poset alphabet with at least two elements. The quantity that governs the answer is the number of letters plus the number of nonempty downsets, and not the alphabet’s cardinality alone: the two-letter chain and the two-letter antichain already separate.

The mechanism behind the sharpness is explicit. Each proper nonempty downset can be activated as a separator by a terminal-letter guard, because a letter outside a downward-closed set cannot be absorbed by the block that repeats it. The full downset admits no such letter and can be activated only after an earlier proper downset has supplied a guard of positive limit length, which no finite initial segment of a periodic block can accommodate. When no earlier proper downset is available the activation genuinely fails, and the universal-only family has maximal order type $H_n \cdot \omega$ rather than H_{n+1} ; the singleton alphabet is the same phenomenon.

The proposed solution is therefore not merely a numerical lower bound matching a known upper bound. It identifies and isolates the structural mechanism responsible for the sharpness, in a form that is stated for arbitrary finite posets, classified for arbitrary allowed families of periodic tails, made effective by a canonical normal form and a linear-time symbolic comparator, and audited both by worked negative controls and by a reproducible test suite. Its external status remains that of an unrefereed proposed solution, in the sense set out in Section [23](#).

A A compact self-contained statement of the construction

This appendix restates the lower-bound mechanism on one page, in the all-guarded form of Theorem 6.5 and Remark 6.6. Fix a nonempty finite poset P , an allowed family $\mathcal{G} \subseteq \mathcal{J}(P) \setminus \{\emptyset\}$, and a new nonempty downset $D \notin \mathcal{G}$ with $D \not\subseteq E$ for every $E \in \mathcal{G}$. Set

$$g_D(u) = \begin{cases} uc, & D \neq P, \quad c \in P \setminus D, \\ ucB_R, & D = P, \quad R \in \mathcal{G} \text{ proper}, \quad c \in P \setminus R. \end{cases}$$

The second line is defined only when the required R exists; that proviso is exactly the exceptional case of Theorem 7.1. For every $m < \omega$ put

$$\Phi_m(\mathbf{u}) = g_D(u_0)B_D g_D(u_1)B_D \cdots B_D g_D(u_m).$$

Then

$$\Phi_m(\mathbf{u}) \preccurlyeq \Phi_m(\mathbf{v}) \iff (\forall i \leq m) u_i \preccurlyeq v_i.$$

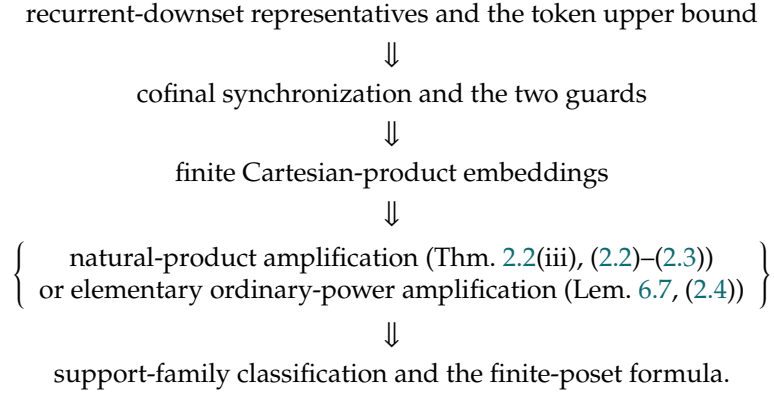
The proof is synchronization (Lemma 4.4 with Corollary 4.5), plus nonleakage (Lemma 6.2), plus cancellation (Lemmas 5.1 and 6.1). If $o(\mathcal{W}(P, \mathcal{G})) = H_q$ with $q = n + |\mathcal{G}|$, the equivalence yields

$$H_{q+1} = \sup_{m < \omega} H_q^{m+1} \leq o(\mathcal{W}(P, \mathcal{G} \cup \{D\})) \leq H_{q+1},$$

where the left equality may be read either through the natural powers of (2.2)–(2.3) or through the ordinary powers of (2.4) and Lemma 6.7, and the right inequality is Proposition 3.8. This is the complete lower-bound mechanism used in the article.

B A compact proof-dependency map

In outline, the argument has five layers:



The fourth layer is the only one at which the article offers two independent routes; either suffices for Theorem 1.1. The table below expands the same dependencies statement by statement.

Statement	Inputs	Role
Finite representation, Proposition 3.5	Finiteness of P ; recurrent labels; monotone concatenation	Identifies the full word quotient with a finite-token quotient.
Atom-count upper bound, Proposition 3.8	Finite-word formula; monotone surjection	Gives $o(\mathcal{W}(P, \mathcal{F})) \leq H_{n+ \mathcal{F} }$.
Cofinal localization, Lemma 4.1	Finitely many target tokens; strict increase; periodic recurrence	Finds the target block used cofinally by each source marker.
Synchronization, Lemma 4.4	Cofinal localization; no old block contains the new downset	Forces the i th new marker to correspond to the i th new marker.
Proper product embedding, Theorem 5.2	Synchronization; a letter outside the proper downset	Embeds every finite Cartesian power of the old stage.
Limit padding, Lemma 6.1	A proper old block; an excluded letter; final-letter cancellation	Produces an order-reflecting map into nonzero limit lengths.
Universal product embedding, Theorem 6.3	Synchronization; limit padding; the finite-bound contradiction	Adds the universal block without losing the product lower bound.
Family classification, Theorem 7.1	The two product maps; classical natural product; atom upper bound	Computes every selected family, including its sole exception.
Main theorem, Theorem 1.1	Full family has a proper downset when $n \geq 2$; finite representation	Gives $\omega^{\omega^{n+j(P)-2}}$ and the two boundary cases.
Unified product embedding, Theorem 6.5	The same synchronization, with one case-split guard map	Restates Theorems 5.2 and 6.3 as one equivalence.
Elementary amplification, Lemma 6.7 and Corollary 6.8	A maximal linear extension; the last-unequal-coordinate order	Replaces the natural-product import in the main line.
One-block normal form, Lemma 3.4	Last occurrence outside the recurrent downset	Supplies the admissible prefixes used by (20.3).
Equal-count rigidity, Lemma 20.2	Cofinal localization at a fixed block count	Whole-block containment; used only in Sections 20 and 21.3.
Exact-length strata, Propositions 20.1 and 20.3	Equal-count rigidity; natural sum and product	Supplementary; yields the warning (20.7).
Canonical representatives, Proposition 21.2	Normalized prefixes; equal-count rigidity; (20.3)	Makes the finite representation unique and effective.

C Minimal worked use of the code

The following example can be run with `code/` on the Python import path. Downset masks 1, 2, and 3 represent $\{a\}$, $\{b\}$, and $\{a, b\}$.

```
from omega2 import (FinitePoset, letter, omega, embeds, normalize,
                    encode_product, theorem_value)

P = FinitePoset.antichain(2)
a, b = letter(0), letter(1)
A, B, U = omega(1), omega(2), omega(3)

assert embeds(P, (a, U), (U,)) # finite-prefix absorption
assert embeds(P, (A,), (U,)) # pure tail into universal tail
assert not embeds(P, (A, B), (U,)) # two cofinal blocks cannot share one
assert not embeds(P, (U,), (A, B)) # no eligible cofinal destination

assert normalize((a, U)) == (U,) # canonical quotient representative
assert normalize((b, A)) == (b, A) # b is outside the downset of A

# The unified guarded product map, introducing B over the family {A}.
print(encode_product(P, (1,), 2, (((), (a,))),))
# (('f', 0), ('w', 2), ('f', 0), ('f', 0))

print(theorem_value(P))
# omega^(omega^4)

C = FinitePoset.chain(2)
print(theorem_value(C))
# omega^(omega^3)

print(theorem_value(P, family=(P.full,)))
# omega^(omega^1 + 1) = omega^(omega + 1)
```

The last output illustrates the exceptional universal-only family. Its maximal order type is much smaller than $H_3 = \omega^{\omega^2}$, even though the two finite letters and the universal block form three distinct formal tokens.

D Literature and reproducibility record

The chosen source was checked both in arXiv HTML and in the PDF rendering; Example 3.15 is on printed page 10, which is zero-based PDF page 9. Its version identifier is 2409.03199v2. The source of the open question, the two standard maximal-order-type references, and the distinction between proven classical inputs and the present proposed argument are recorded in `notes/literature_audit.md`.

The archive contains this source and its compiled PDF, a reusable `references.bib`, the symbolic module, the reproducible verification script, machine-readable test results, example and downset-count tables, a proof-audit note, and a README. It contains no third-party article PDFs, font files, checksum files, or LaTeX build intermediates. The file inventory is the following.

File	Purpose
<code>article.tex</code>	Complete editable source of this manuscript.
<code>article.pdf</code>	Compiled article.
<code>references.bib</code>	Bibliographic records, maintained alongside the embedded <code>thebibliography</code> .
<code>build.sh</code>	Three-pass <code>pdflatex</code> build script.
<code>README.md</code>	Scope, result list, reproduction, and library usage.

File	Purpose
code/omega2.py	Finite posets, downsets, exact token comparators, normalization, the guard maps, and the product encodings.
code/verify.py	Exhaustive and seeded checks, negative controls, and table generation.
data/verification.json, data/verification.txt data/test_summary.tex data/examples.csv data/ideal_counts.csv	Recorded output of the default verification run.
data/finite_posets.csv	Test-count macros used by this article. The computed finite-poset examples of Table 1. Downset-count histogram for naturally labelled posets of size at most five.
notes/proof_audit.md notes/literature_audit.md	The 50 naturally labelled posets through size four, individually, with predecessor and successor masks, downset counts, and the exponent predicted by (1.1). Hypotheses, critical steps, and known false shortcuts. Source, version, and literature-search scope.

E The finite-alphabet Higman formula

This section records a proof of the finite case of (2.1), using Higman’s wqo theorem and the standard calculus (10.2). It also explains why the separator construction in the main argument has the right ordinal shape.

This proves clause (v) of Theorem 2.2, rather than asserting a new result.

Proof. Induct on n . For $n = 1$, words are ordered by their finite lengths, so the type is $\omega = H_1$. Assume $n \geq 2$ and the result for smaller alphabets.

Lower bound. Choose a maximal letter $b \in R$ and set $Q = R \setminus \{b\}$. In words

$$u_0 b u_1 b \cdots b u_m, \quad u_i \in Q^*,$$

every source b must map to a target b , since $b \not\leq q$ for $q \in Q$. The number of displayed b ’s cannot decrease under embedding. When the two words have the same number, all displayed b ’s match in order, forcing the intervening coordinates into their corresponding intervals. Consequently, the layer with m occurrences of b is an induced copy of $(Q^*)^{m+1}$. The layers are disjoint and their index is monotone, so lemma 10.1, with $o(Q^*) = H_{n-1}$, gives lower bound H_n .

Upper bound. Fix a forbidden nonempty word $w = p_1 \cdots p_\ell$. A word that does not contain w as an embedded subsequence has a greedy factorization

$$u_1 b_1 u_2 b_2 \cdots u_j b_j u_{j+1}, \quad 0 \leq j < \ell,$$

where $b_i \geq p_i$ and u_i uses only letters in $Q_i = R \setminus \uparrow p_i$. Each Q_i is a proper subalphabet. This factorization is obtained by greedily matching p_1 , then p_2 , and so on, stopping when the next letter cannot be matched.

For a fixed j and fixed matched letters b_i , concatenation is a monotone surjection from a product of $j + 1$ orders Q_i^* onto this part of the residual. By induction every nonempty Q_i^* has type at most H_{n-1} ; an empty Q_i contributes the one-element order, which also satisfies the bound. There are only finitely many choices of j and of the b_i . Thus the residual $R^* \setminus \uparrow w$ is a monotone image of a finite disjoint union of products, whose maximal order type is bounded by a finite natural sum of ordinals at most

$$H_{n-1}^{\otimes \ell} = \omega^{\omega^{n-2} \cdot \ell}.$$

Every such finite sum is strictly less than H_n . The residual for the empty forbidden word is empty. Using the residual identity in (10.2) and the fact that H_n is a limit ordinal gives $o(R^*) \leq H_n$. $\square$

The transfinite proof differs in one crucial respect. A new separator is an infinite word, not a single letter. Old coordinates can in principle enter its beginning even when its occurrences have all been matched. Protected tails are what restore the coordinate conclusion.

F Exact symbolic embedding below ω^2

The test program represents a word as

$$p_1 c_1^\omega p_2 c_2^\omega \cdots p_t c_t^\omega q, \quad (\text{F.1})$$

where each p_i, q is a finite word and each c_i is a finite nonempty cycle. This is an exact symbolic representation of a transfinite word, not a truncation. It is sufficiently general to represent every class below ω^2 over a finite alphabet, by the normal-form argument, although the tests use bounded finite parameter sets.

F.1 The greedy algorithm

Maintain a cursor in the target. Within a target block it consists of its block index and a finite offset in its finite prefix or periodic tail. After the final infinite block it is an offset in the finite suffix. Process a source finite prefix one letter at a time, always choosing the earliest target position carrying a label above the source label. In a periodic tail, checking one cycle suffices to find the earliest match; if no cycle letter is suitable, skip to the next target block.

For a source periodic tail c^ω , find the earliest target infinite block, at or after the cursor, whose recurring cycle d satisfies

$$(\forall a \text{ occurring in } c)(\exists b \text{ occurring in } d) a \leq_p b. \quad (\text{F.2})$$

If there is no such block, embedding fails. Otherwise embed the periodic tail cofinally into that block and advance the cursor to the beginning of the next target block. Continue with the next source prefix. After all source infinite blocks are consumed, greedily process the finite source suffix.

F.2 Why the algorithm is exact

Greedy matching of a finite letter is safe because replacing its target position by an earlier suitable one can only enlarge the remaining target interval. This is the usual finite-subsequence argument and applies unchanged inside a transfinite target.

An infinite periodic source tail has infinitely many occurrences of every label in its cycle. If it embeds into a finite concatenation of target infinite blocks and a finite suffix, its image ends cofinally in one target infinite block. A source tail of the same periodic type therefore embeds into that block's recurring tail. This forces (F.2); finite exceptional target prefixes cannot supply infinitely many occurrences of a missing label.

Conversely, (F.2) suffices: place successive source letters in suitable positions of successive sufficiently late target cycles. Any finite cursor offset is harmless. Choosing the earliest compatible target block leaves the largest possible collection of whole target blocks for subsequent source material. No subsequent source position could remain in the same target block after a cofinal image. Thus advancing to the next block is both sound and optimal. Induction over the finitely many source blocks proves correctness.

F.3 Test families and negative controls

The marker tests compare all finite binary source words of length at most four with all finite binary target words of length at most four, for both the chain and antichain letter orders. There are $2(1+2+4+8+16)^2 = 1,922$ pairs.

The equal-block tests use up to four exact ω -blocks, with finite prefixes and one of the cycles 0, 1, or 01. The separator tests first add 1^ω to a language whose infinite blocks have cycle 0. For the antichain they also add $(01)^\omega$ to a language whose infinite blocks are monochromatic. Both equal and unequal separator counts are tested, including positive comparisons and self-comparisons.

As negative controls against an incorrectly rigid concatenation implementation, the program verifies the genuine absorption $aa^\omega \equiv a^\omega$. It also checks that over a singleton alphabet, the embedding decision agrees exactly with comparison of the ordinal lengths $\omega t + s$ on the finite test range.

References

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